

## APPRENTICE'S DAILY DIARY

### UNDERGRADUATE / DIPLOMA / NVQ 4 / 5 / 6 / CERTIFICATE

Name : A.G.C.O. Atadaswala.

Apprentice's Private Address : Poramadulla, Riki Nagaskada.

Contact Phone Number : 070 700 7668

Category of Apprenticeship : Undergraduate, Bachelor of Science (Hons)

Field of Engineering : Mechanical Engineering

Registration Number given by the University / ~~Institute~~ / College : 200055B

Registered Number of the Contract : UGC/UOM/ENR / 00365 / 2023 (2020)

Overall Period of Training : 24 weeks From : 27/11/2023 To : 13/05/2024

## NOVATION

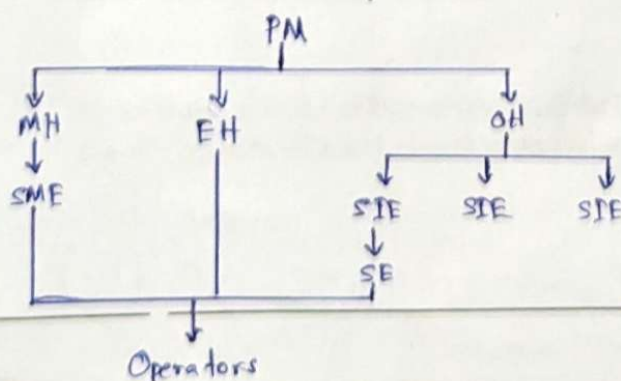
| Name and Address of Establishment                                                                     | Workshops / Worksites                              | Period     |            | Signature of Officer In Charge                                                        | Designation |
|-------------------------------------------------------------------------------------------------------|----------------------------------------------------|------------|------------|---------------------------------------------------------------------------------------|-------------|
|                                                                                                       |                                                    | From       | To         |                                                                                       |             |
| Lakdhanavi Ltd.<br>Yugadanari 300MW CCPP,<br>Muthurajawela Industrial<br>Zone, Kerawalapitip, Wattala | Yugadanari 300MW<br>combined cycle<br>Power Plant. | 27/11/2023 | 30/05/2024 |  | ME          |
|                                                                                                       |                                                    |            |            |                                                                                       |             |
|                                                                                                       |                                                    |            |            |                                                                                       |             |
|                                                                                                       |                                                    |            |            |                                                                                       |             |
|                                                                                                       |                                                    |            |            |                                                                                       |             |
|                                                                                                       |                                                    |            |            |                                                                                       |             |

WEEK NO: 01

| FOR THE WEEK ENDING      |       | TRAINING LOCATION                                                                                  |
|--------------------------|-------|----------------------------------------------------------------------------------------------------|
| Sunday... 03 / 12 / 2023 |       | Yugadanavi Power Plant<br>Kerawalapitiya.                                                          |
| DAY                      | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                          |
| Monday                   | 11/27 | Visited to the Yugadanavi Power Plant and got some knowledge about safety.                         |
| Tuesday                  | 11/28 | Field visit was done and we had brief idea about the systems and processes.                        |
| Wednesday                | 11/29 | We had our first task to research about fire hydrant boxes and got an idea to seal them.           |
| Thursday                 | 11/30 | Visited to the fuel oil treatment plant and got some knowledge about its processes and the system. |
| Friday                   | 12/01 | Researched the manual of the agitator in FOTP and made a checklist.                                |
| Saturday                 | 12/02 | Holiday                                                                                            |
| Sunday                   | 12/03 | Holiday                                                                                            |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

Hierarchy of the power plant



PM - plant manager  
 MH - mechanical head  
 EH - electrical head  
 OH - operational head  
 SME - senior mechanical head Engineer  
 SIE - shift interchange engineer  
 SE - shift engineer



## The FOTP Process

There are two gas turbines in this power plant. Mainly heavy fuel oil (HFO) is used as the fuel. Due to its low quality, so many contaminants like Na, K, V, Cd...etc. are mixed with HFO. To increase durability of turbines and other parts, these contaminants are removed in FOTP. This process is called as fuel oil treatment process.

Initially, 60°C untreated HFO is transferred to the FOTP from untreated tanks. Then the temperature of this oil is increased into 90°C using two heat exchangers. Treated HFO is used in first heat exchanger and hot water is used in second heat exchanger.

Next, demulsifier is mixed with the oil in the mixer A and contaminants are separated in the separator A. Again this is repeated in mixer B with separator B. After that the line is finished and the slug is removed. The condition of the oil is tested and if it is passed, the oil is sent to the treated tanks. Otherwise, it is sent again to the untreated tanks. The waste water, slug and some oil is transferred to two tanks and all of the oil is removed in that tanks. This oil is sent to the untreated tanks and waste water is sent to the CMB. ETP. Then oil was again removed and send to the untreated tanks. Water is send to the CMB.

Chalangann  
SIGNATURE OF TRAINEE

### REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

|  |  |  |
|--|--|--|
|  |  |  |
|  |  |  |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC. DIMENSIONS AND SKETCHES TO BE GIVEN WHERE NECESSARY

A checklist was prepared regarding the work carried out in the power plant. There are two heat exchangers used in the power plant. From them the hot plate heat exchanger is used to increase the temperature of HFO. The other heat exchanger is used to cool the HFO. The oil is transferred to the treated tanks. The waste water, slug and some oil is transferred to two tanks and all of the oil is removed in that tanks. This oil is sent to the untreated tanks and waste water is sent to the CMB. ETP. Then oil was again removed and send to the untreated tanks. Water is send to the CMB.

DATE: 02/01/2023

W.P.D.N.N. Wijesinghe  
Mechanical Engineer  
DESIGNATION AND SIGNATURE



WEEK NO: 02

FOR THE WEEK ENDING

TRAINING LOCATION

Sunday 10/12/2023

Yugadanavi Power Plant  
Kerawalapitiya

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                                            |
|-----------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | 12/04 | Prepared a checklist for brazed plate heat exchanger according to the owner's manual.                                                                |
| Tuesday   | 12/05 | Visited to the unloading area to get some knowledge about the processes and systems. Also went to the FOTP to understand the heat exchanger process. |
| Wednesday | 12/06 | Visited to the gas turbine and observed about its first stage, outer side processes like pre heating.                                                |
| Thursday  | 12/07 | Rechecked about some checklists made excels for them. Visited to the tankyard to understand about that area.                                         |
| Friday    | 12/08 | Observed about preventive maintenance of motors and gear boxes in water treatment plant.                                                             |
| Saturday  | 12/09 | Holiday                                                                                                                                              |
| Sunday    | 12/10 | Holiday.                                                                                                                                             |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

A checklist was prepared regarding heat exchanger.

There are many heat exchangers used in this power plant. From them, two brazed plate heat exchangers are used in FOTP. The first heat exchanger is used to increase the temperature of untreated HFO from 60°C. In this process, treated HFO is used. The other heat exchanger is used to increase temperature of previous untreated HFO into 90°C.



Hot water is used in this process.

Heat exchanger has a number of stainless steel plates brazed together to form a plate pack. One fluid flows in between two plates and the other one flows in the opposite side of those two plates. Due to temperature difference of these fluid, temperature is transferred from one to other.

For a plate, there are two gaskets in both sides. These gaskets are used to seal these plates without leaking. There are two inlets and two outlets and clamping bolts are used to seal all the plates equally.

After understanding all of these things, a checklist was made according to the owner's manual.

According to the owner's manual, preventive maintenance is done after 6 months. Following are the things should be considered after a preventive maintenance.

- \* Check whether all plates are cleaned.
- \* Check whether all gaskets are fitted well.
- \* Check whether all clamping bolts are tightened equally.
- \* Check whether all inlets and outlets are fitted well.

Chalangana  
SIGNATURE OF TRAINEE

#### REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

DATE : 21/01/2023

W.P.D.N.N. Wijesinghe  
Mechanical Engineer  
DESIGNATION AND SIGNATURE



WEEK NO: 03

FOR THE WEEK ENDING

Sunday 17 / 12 / 2023

TRAINING LOCATION

Yugadanavi Power Plant  
Kesawalapitiya.

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                                                                                                                                       |
|-----------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | 12/12 | Checked the preventive maintenance of some machines used in the power plant and created checklists for them.                                                                                                                                    |
| Tuesday   | 12/12 | Visited to the auxiliary boiler, where we gained knowledge about its process.                                                                                                                                                                   |
| Wednesday | 12/13 | Visited the water treatment plant, where numerous pneumatic valves and filters are utilized. We gained knowledge about the chemicals used to reduce the conductivity of seawater and how conductivity is controlled through mechanical systems. |
| Thursday  | 12/14 | There was a damage in low pressure drum inlet valve motor of the heat recovery steam generator. Studied about the HRSG process and the motor repair procedures.                                                                                 |
| Friday    | 12/15 | Gained knowledge about preventive maintenance in couplings of motors in auxiliary boiler.                                                                                                                                                       |
| Saturday  | 12/16 | — Holiday —                                                                                                                                                                                                                                     |
| Sunday    | 12/17 | — Holiday —                                                                                                                                                                                                                                     |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

### Heat Recovery Steam Generator (HRSG)

The HRSG is a water tube boiler system used to generate steam using exhaust gas from a gas turbine. There are two pressure sides in HRSG. Low pressure and high pressure.



The high pressure (HP) side is used to generate electricity at high pressure, while the low pressure (LP) side is used to generate electricity in low pressure turbine. The main components on each pressure side include the economizer, evaporator and super heater. Additionally, there is a drum for both LP and HP.

The process :-

The condensated water first travels through the LP economizer, with a Condensate Extraction Pump used to pump LP water into the LP economizer. Subsequently, this water travels through the LP drum, then passes through the LP evaporator once again before returning to the LP drum. The discharged water from the LP drum then travels through the economizer, and the steam generated in the LP drum moves through the LP superheater. These economizer, evaporator and superheater are constructed from tubes, increasing the contact surface area with exhaust gases. The LP superheated steam is employed to generate electricity using the LP steam turbine, and the LP pressure is approximately 7 bars.

The HP steam process follows a similar pattern. A Boiler Feed Water Pump is utilized to raise the pressure of LP water to around 90 bars. The LP water from the LP drum is pumped through the HP economizer and then into the HP drum.



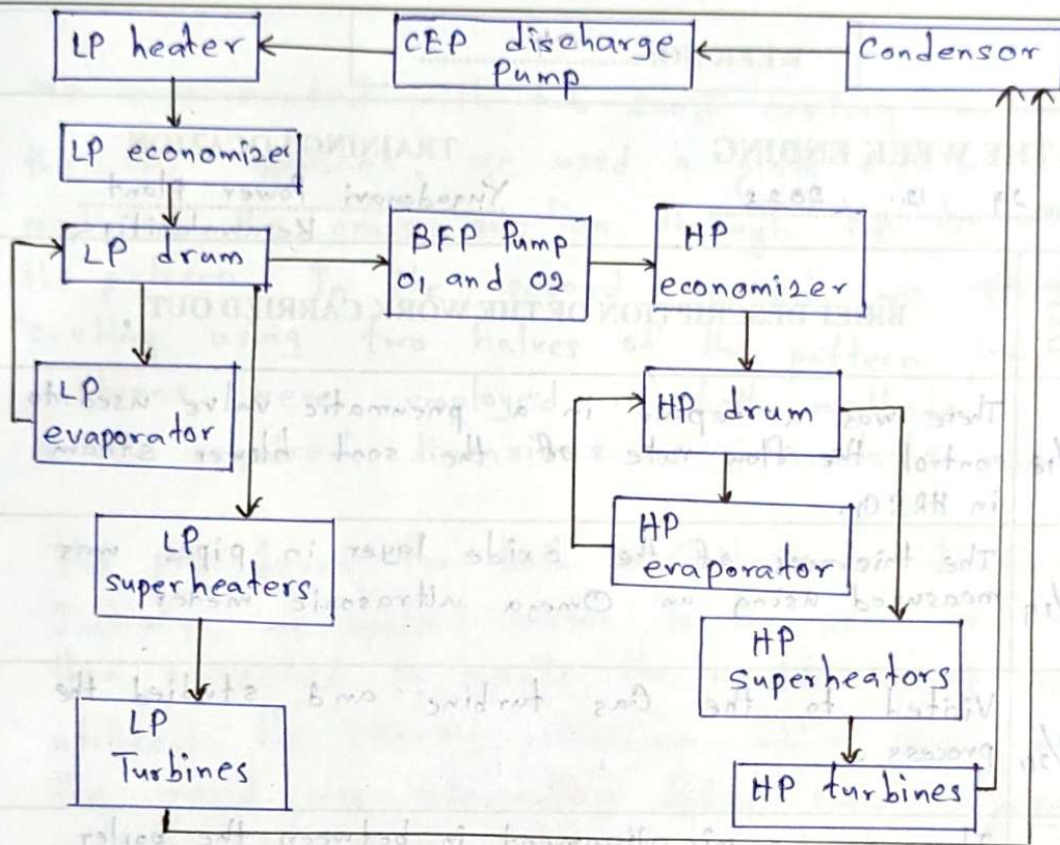
The subsequent process mirrors that of the LP side. The superheated HP steam is used to generate electricity using the HP steam turbine, with a temperature of around 500 degrees Celsius for the superheated HP steam.

There is a blowdown tank for each HRSG, which is used to remove water from the LP and HP drums. A line from the LP drum is connected to the blowdown tank, referred to as instant blowdown. Two lines are connected from the HP drum to the blowdown tank, known as instant blowdown and continuous blowdown. Water discharges continuously from the CBD.

### Soot Blower System :-

Due to the flue gas, carbon is deposited on the outer surface of the HRSG tubes, which can reduce the efficiency of the steam generation process. Therefore, soot blowers are used to remove this deposited carbon. There are 10 soot blowers, and each soot blower takes seven minutes to complete its work. Two limit switches are used to control each soot blower. A pressure of 90 bars superheated steam is used in the soot blowers. Additionally, there is a pneumatic flow control valve to regulate the flow of steam for the soot blower.





Chalangana

SIGNATURE OF TRAINEE

REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

DATE : 12/01/2024

W.P.D.N.N. Wijesinghe (M.E.)  
DESIGNATION AND SIGNATURE



WEEK NO: 046

FOR THE WEEK ENDING

TRAINING LOCATION

Sunday 24 / 12 / 2023

Yugadanari Power plant  
Kerawalapitiya.

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                             |
|-----------|-------|---------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | 12/18 | There was a repair in a pneumatic valve used to control the flow rate of the soot blower steam in HRSG.                               |
| Tuesday   | 12/19 | The thickness of the oxide layer in pipes was measured using an Omega ultrasonic meter.                                               |
| Wednesday | 12/20 | Visited to the Gas turbine and studied the process.                                                                                   |
| Thursday  | 12/21 | There was a misalignment in between the boiler feed water pump and its cou motor. The Easy-laser XT660 device was used to align this. |
| Friday    | 12/22 | A fireline pipe endcap was designed according to the dimensions using solidworks and it was casted using sand casting methods.        |
| Saturday  | 12/23 | - Holiday -                                                                                                                           |
| Sunday    | 12/24 | - Holiday -                                                                                                                           |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

Casting of a fireline pipe endcap.

We began manufacturing fireline endcaps to meet the requirements of our power plant, utilizing aluminium and employing the sand-casting method.

We experimented with two sand casting methods. In the first approach, we used a plate with a hole, matching the cross-section through the centerline of the pattern. In the second method, we attempted casting using two halves of the pattern. Wooden patterns were employed in both methods, crafted to the required dimensions using a lathe machine.

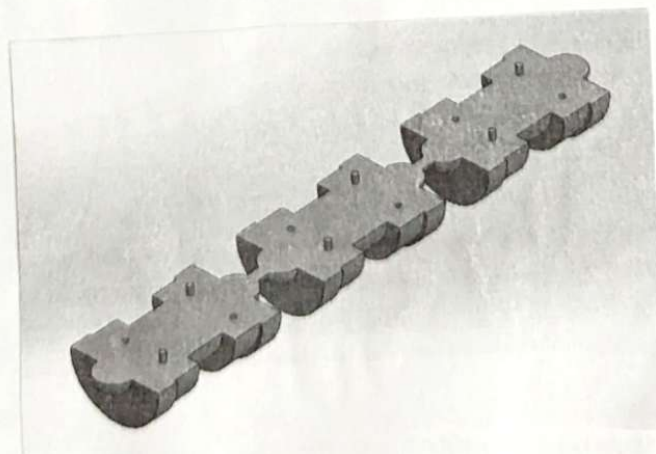
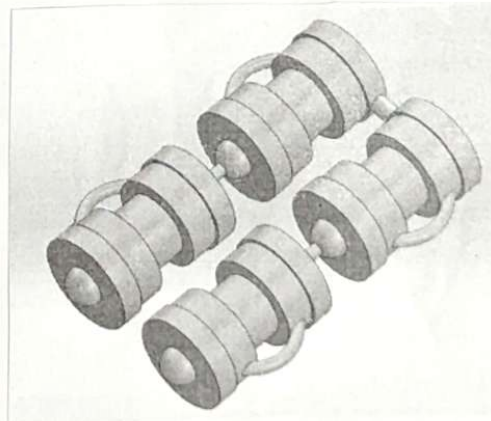
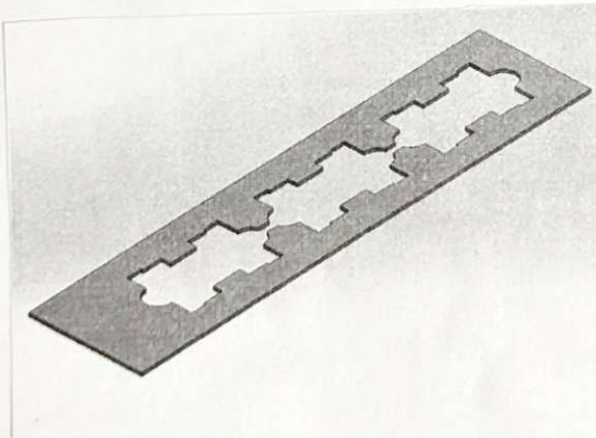
The procedures for both methods were identical. Initially, we applied water to the sand and mixed it, then proceeded to create the mould. Sprues were utilized for pouring aluminium, while risers ensured the mould was adequately filled. Gate systems, runners and vent holes were added as necessary. Additionally, a core was prepared separately and incorporated into the mould.

Discarded aluminium found in the plant was melted using a furnace. Subsequently, the molten aluminium was poured into the mould until it overflowed through the risers. After a few minutes, the mould was removed, and the component was cleaned. Machining was employed for achieving a superior surface finish.



## Designed Patterns using SolidWorks.

I had another task to design a pattern for those end caps and using SolidWorks for 3D Printing. Initially, I gathered all the necessary dimensions and drew a plan view and a front view of the pattern. Subsequently, I designed a plate with three series of holes and three series of patterns, all tailored to the required size. This plate is employed for manufacturing end caps using the previously mentioned first method. Additionally, I created four more patterns, each with two halves, specifically designed for the second method.





Chalamana

SIGNATURE OF TRAINEE

REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

- Holiday -

- Holiday -

DATE: 28/01/2024

W.P.D.N.N. Wisesinghe (HE)

DESIGNATION AND SIGNATURE



WEEK NO: 05

FOR THE WEEK ENDING

Sunday 31 / 12 / 2020

TRAINING LOCATION

Yugadanari Power Plant  
Kerawalapitiya.

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                                                             |
|-----------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | 12/25 | - Christmas day -                                                                                                                                                     |
| Tuesday   | 12/26 | - Poya day -                                                                                                                                                          |
| Wednesday | 12/27 | Had to check the preventive maintenance of some machines and created checklists for them. Subsequently, I assisted in uploading these checklists to the plant system. |
| Thursday  | 12/28 | Had to create another checklists for another machines and uploaded them in the plant system.                                                                          |
| Friday    | 12/29 | Again Visited to the gas turbine and fully understood about its process and also lubrication process.                                                                 |
| Saturday  | 12/30 | - Holiday -                                                                                                                                                           |
| Sunday    | 12/31 | - Holiday -                                                                                                                                                           |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

### Gas Turbine

In this plant, two gas turbines and one steam turbine are used. Gas turbines operate in an open-cycle process. Each gas turbine can generate around 100 MW. Heavy Fuel Oil (HFO) and Diesel are used as the main fuels in the gas turbines.

Outside the Gas Turbine (GT) Plant, there are two filters for each Diesel Oil (DO) and Heavy Fuel Oil (HFO) line. One is used for operation and the other serves as a standby. The pressure difference between these filters is monitored to check their condition. If this pressure difference is higher, the filter is switched to the standby one.

Sulphur can mix with the fuel oil, affecting the nozzles. Therefore, Vanadium is used to bond with sulphur in a vanadium skid. After passing through the filters, the fuel is transferred to this vanadium skid. A three-way valve is installed after the vanadium skid, controlling the direction of HFO or DO to the GT. During startup and shutdown, DO is used. After 20 minutes of startup, the DO line is switched to the HFO line. Normally, at this point, the GT can produce around 10 MW.

The temperature of the turbine shaft increases when the GT is running. Therefore, if the GT is not generating electricity, the shaft rotates at a low rpm. If not, it can buckle. A turning gear motor is used to rotate the shaft in low rpm.

Shaft rpms :-

- Turning gear motor - 120 rpm
- Cranking motor - 780 rpm
- Fire - 3000 rpm.



There are three types of lube oil pumps to lubricate the bearings and lift oil pumps to elevate the shaft around 3-4 mm. The three types of lube oil pumps are AC pump, DC pump and mechanical pump. The pressure of the lift oil is around 200 bars. The AC lube oil pump is used upto 1800 rpm of shaft speed. After that, the pump is changed to the mechanical pump.

The temperature of the oil return line of the lube oil is very high due to the higher temperature of the bearings. Consequently, there can be mist in the return line. A mist eliminator is used to prevent this. The mist eliminator contains a cartridge that removes vacuum and separates oil. The oil then goes back to the lube oil sump.

There are mainly four compartments in the AT. They are auxiliary, turbine, load and generator.

**Auxiliary compartment :-**

After the previous three-way valve, the HFO is transferred to this compartment. The HFO is then passed through a high-pressure fuel filter. Following that, a flow divider is used to split the flow into 14 lines, which are employed to transfer oil into 14 turbine nozzles. Lube oil pumps, mist eliminator, turning gear motor and cranking gear motor are all located in this compartment.

### Turbine Compartment:-

Compression and the combustion are the two parts in this compartment. The compressor has 17 stages of axial flow compressors with variable inlet guide vanes (IGV). The compression ratio is 12:1. Each stage consists of a set of rotating blades and a set of stationary blades. Among these 17 stages, 15 are individual wheels and 2 are stub shafts.

In the combustion section, there are 14 nozzles with 14 check valves for each one. During startup, each valve is initially closed and fuel is returned. Then the check valve is opened, allowing oil to pass through. This is known as purge.

Two spark plugs are present for ignition. In the turbine, compressed air and atomized oil are used for firing.

### Load Compartment:-

In load compartment, the exhaust is transferred to the HRSG.

### Generator Compartment:-

In this compartment, the shaft is connected to the generator. That is used to generate electricity. Two lift oil pumps are also located in this compartment.



| FOR THE WEEK ENDING                                                                                                                                                                                                                                                                                                                                                                                   |       | TRAINING LOCATION                                                                     |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------|--|
| Sunday of 01 2024                                                                                                                                                                                                                                                                                                                                                                                     |       | Tandapani Power plant<br>Karnataka                                                    |  |
| DATE                                                                                                                                                                                                                                                                                                                                                                                                  | D.V.R | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                             |  |
| 01                                                                                                                                                                                                                                                                                                                                                                                                    | 01    | I had to update the plant system with previously prepared checklists.                 |  |
| 02                                                                                                                                                                                                                                                                                                                                                                                                    | 02    | I had to update the plant system with previously prepared checklists.                 |  |
| 03                                                                                                                                                                                                                                                                                                                                                                                                    | 03    | We visited the steam turbine plant and joined inside into the processes and systems.  |  |
| 04                                                                                                                                                                                                                                                                                                                                                                                                    | 04    | Inspected all the hydraulic boxes to compile a list of components inside those boxes. |  |
| <p align="center"><u>Chalanga</u><br/>SIGNATURE OF TRAINEE</p>                                                                                                                                                                                                                                                                                                                                        |       |                                                                                       |  |
| REMARKS AND CERTIFICATION BY THE ENGINEER / T.O                                                                                                                                                                                                                                                                                                                                                       |       |                                                                                       |  |
| 05                                                                                                                                                                                                                                                                                                                                                                                                    | 05    | - Holiday -                                                                           |  |
| 06                                                                                                                                                                                                                                                                                                                                                                                                    | 06    | - Holiday -                                                                           |  |
| <p>DETAILS AND NOTES OF WORK CARRIED OUT PROBLEMS ENCOUNTERED AND HOW SOLVED ETC. DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE</p> <p align="center">About Steam Turbine Plant</p> <p>In a combined cycle power plant, a steam turbine is used to generate electricity using exhaust gas from a gas turbine. The water in the heat exchanger is heated and produces high pressure steam.</p> |       |                                                                                       |  |
| DATE : 20/01/2024                                                                                                                                                                                                                                                                                                                                                                                     |       | <p align="center"><u>P. D. N. N. Sangeetha</u> (ME)<br/>DESIGNATION AND SIGNATURE</p> |  |

WEEK NO: 06.

## FOR THE WEEK ENDING

Sunday 07/01/2024

## TRAINING LOCATION

Yugadanari Power Plant  
Kerawalapitiya.

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                                                       |
|-----------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | 01/01 | I had to update the plant system with previously prepared checklists.                                                                                           |
| Tuesday   | 01/02 | I had to update the plant system with previously prepared checklists.                                                                                           |
| Wednesday | 01/03 | We visited the steam turbine plant and gained insights into the processes and systems.                                                                          |
| Thursday  | 01/04 | Inspected all fire hydrant boxes to compile a list of components inside those boxes.                                                                            |
| Friday    | 01/05 | A fault was discovered in cooling tower fan 22, where old grease had structured. We learned the process of cleaning the bearings and adding new grease to them. |
| Saturday  | 01/06 | - Holiday -                                                                                                                                                     |
| Sunday    | 01/07 | - Holiday -                                                                                                                                                     |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

### About Steam Turbine Plant.

In a combined cycle power plant, a steam turbine is used to generate electricity using exhaust gas from a gas turbine. The water in the Heat Recovery Steam Generator absorbs the temperature from the exhaust gas and produces High-Pressure



and Low - pressure steam. The steam generated in the HRSG is then directed to the steam turbine, where it converts the thermal energy into mechanical energy by turning a shaft. As the turbine spins, it drives a generator that produces electrical power. This dual-cycle configuration allows the combined cycle power plant to harness waste heat from the gas turbine cycle, improving overall efficiency and reducing environmental impact.

SCS is the type of steam turbine used in the Yugadanari Power Plant. The turbine rotor has total of 20 stages of blades, with LP admission steam being fed at the 15<sup>th</sup> stage, and extraction taken from the 17<sup>th</sup> stage. There are two pressure sides: low pressure and high pressure. The pressure on the low pressure side is approximately 7 bars, while on the HP side, its value is around 90 bars. The temperature of the LP steam is approximately 220 degrees Celsius, and the temperature of the HP steam is around 500 degrees Celsius.

These two turbines are connected to a single shaft to ensure that the rotational speed remains the same. Otherwise, the frequency of the generated electricity could vary.

The steam turbine consists of three parts, the condenser, turbine and generator. The condenser is used to convert all the steam into water because steam cannot be pumped. This condenser has two tanks, each with inlet and outlet lines for sea water. The inner line of the condenser is made from titanium for its durability. Following the condensation process, two condensate extraction pumps are utilized to pump the water back to the Heat Recovery Steam Generator.

The generator is connected to the shaft is used to generate electricity. The output voltage of the generator is 14.5 kV. It is a two pole three phase 50 Hz motor. The generator can produce maximum power around 110 MW.

There are four main bearings in the shaft. The generator and the turbine are placed between them. There is a lube oil system to lubricate the bearings. Two AC lube oil pumps are used to pump oil to the bearings. One is working and the other one is standby. For emergency, there is a DC lube oil pump. The temperature of the shafts increases when the ST is running. Therefore, if the ST is not working, the shaft rotates at a low rpm. If not, it can buckle. Rotational speed of this turning gear is 120 rpm.



FOR THE WEEK ENDING

Sunday 12 01 2024

TRAINING LOCATION

Indraprastha Power Plant  
Bachchanpally

BRIEF DESCRIPTION OF THE WORK CARRIED OUT

DAY DATE

12/01/2024

12/01/2024

12/01/2024

12/01/2024

12/01/2024

12/01/2024

12/01/2024

12/01/2024

12/01/2024

12/01/2024

REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

Chalangana.  
SIGNATURE OF TRAINEE

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND SOLUTIONS GIVEN WHERE POSSIBLE

Indraprastha Power Plant using the LabExpert device

In motor, there are parts called bearings that

DATE : 20/01/2024

P. D. N. N. Wisingwe (NE)  
DESIGNATION AND SIGNATURE

WEEK NO: 07

FOR THE WEEK ENDING

Sunday 14 / 01 / 2024

TRAINING LOCATION

Yugadanani Power Plant  
Kerawalapitiya

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                                        |
|-----------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | 01/08 | There was a bearing greasing in AC lube oil pump motors and turning gear motor in auxiliary compartment in AT 02.                                |
| Tuesday   | 01/09 | There was a PM in FOTP line 01 mixer A and B. In this, tightness of bolts, width of the helical gears were checked and filled with new lube oil. |
| Wednesday | 01/10 | In WTP, base plate of the acid tank was corroded. The plate was removed and agitator of this tank fixed to another tank.                         |
| Thursday  | 01/11 | The grease level of the bearings were checked in AT 01 auxiliary compartment motors and TK, VA, EF fans motors using LubeExpert SDT970 device.   |
| Friday    | 01/12 | There was a fuel leak in preheater of untreated tank to FOTP line. We tried to identify the leakage after removing the gasket.                   |
| Saturday  | 01/13 | - Holiday -                                                                                                                                      |
| Sunday    | 01/14 | - Holiday -                                                                                                                                      |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

Lubricating bearings in the motors of the Gas Turbine Plant using the LubeExpert device.

In motors, there are parts called bearings that can rub against each other because of the spinning rod. To prevent damage, we use a slippery substance called lubrication to create a thin layer



between these parts. This reduces the rubbing and keeps the parts from getting harmed easily. When things rub together, they create heat, especially in big motors. Lubrication helps to remove this heat, preventing the motor from getting too hot. Having enough lubrication also makes the motor vibration less and be quieter when the parts are spinning.

In the gas turbine, we did lubrication on the turning gear motors, AC lube oil pump motors, and the TK, VA, EF fan motors. To make sure we used the right amount, we checked it with a special tool called as LubeExpert SDT 270 device. This device is made for checking the condition of machines. It listens to the sound they make, specially the high frequency sounds to see if everything is working properly.

In process,

First, the sensor was connected to the ultrasound device. Then, the initial reading was taken in dBμV. Once the reading was taken, LubeExpert displayed a screen with measurement results, representing RMS, max RMS, peak and Crest factor. The grease requirement was determined based on the dBμV reading. Depending on the results and triggered alarms, we identified the need for lubrication.

If the baseline of the bearings is identified, an idea on the bearing condition can be obtained at this stage.

- In case there is no alarm assigned to Lube position, initial reading will be followed by a screen that offers all choices.
- If an alarm is assigned but not triggered lubrication is not needed. The recommendation made by alarm settings can be accepted.
- If an alarm is assigned and triggered lubrication is needed.

For setting the baseline, there are three methods.

- **Comparison :-** When there is more than one bearing of the same type, load and rpm, multiple bearings can be compared. Each bearing is inspected at the same test point and angle. The decibel levels and sound quality are compared. If there are no substantial difference (less than 8 dB), a baseline dB level is set for each bearing.
- **Set while lubricating :-** While lubrication is being applied, the technician can listen until the sound level drops and then begins to rise. At that point, no more lubrication is added and the dB value is used as the baseline. Alternatively, in a system with continuous monitoring, the dB is trended allowing for quantitative measurement before and after lubrication.



- Historical :- Bearing dB levels are obtained from an initial survey and compared 30 days later. If there is little (less than 8 dB) to no change in dB, then the baseline levels are set and will be used for comparison for subsequent inspections.

From these methods, we used the second method. First got the dB value and then applied grease. Then again checked the dB value. If it reduced, the lubrication was done when it nearly stops decreasing.

Chalangana

SIGNATURE OF TRAINEE

REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

DATE : 21/01/21

W.P.N.N. Wicaksono  
DESIGNATION AND SIGNATURE

WEEK NO: 08

FOR THE WEEK ENDING

Sunday 31/01/2024

TRAINING LOCATION

Yugadanari Power Plant  
Kerawalapitiya.

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                                                |
|-----------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | 01/15 | - Tamil Thai Pongal Day -                                                                                                                                |
| Tuesday   | 01/16 | Visited to the HRS G SWAS system and studied about their process and then identified what are the methods they use to check conductivity of drums.       |
| Wednesday | 01/17 | In sea water reverse osmosis, one valve was tightened. The reason is the glands were more tightened. The tightness was reduced while checking the valve. |
| Thursday  | 01/18 | Visited to the HRS G to understand about a temperature system and Eliminox chemical dosing process and its usages.                                       |
| Friday    | 01/19 | Pumps systems in cw system and valves were identified according to a P&ID.                                                                               |
| Saturday  | 01/20 | - Holiday -                                                                                                                                              |
| Sunday    | 01/21 | - Holiday -                                                                                                                                              |

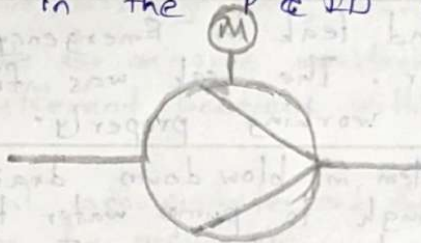
DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

### Identification of pumps in Main cooling water system

There are various types of pumps utilized in this power plant, including submersible pumps, centrifugal pumps, gear pumps, diaphragm pumps and more. Among these, submersible and centrifugal pumps are primarily employed in the Cold End.



According to the P & ID of the main cooling water system, there are main 8 pumps. Two main cooling water pumps, one auxiliary cooling water pump and two tube cleaning circulating pumps. Additionally, there are three pumps for the junction box system. All of these pumps are motor-operated and are designated in the P & ID as depicted in the below.



Among these pumps, the auxiliary cooling water pump is a submersible pump, while the remaining seven pumps are centrifugal pumps.

Chalanpala.

SIGNATURE OF TRAINEE

REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC. DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

Vibration monitoring in Cooling Tower motors.

DATE : 30/01/2024

W. P. D. N. N. Dilipkumar (ME)

DESIGNATION AND SIGNATURE

WEEK NO: ..... 09. ....

| FOR THE WEEK ENDING              |       | TRAINING LOCATION                                                                                                                                                                          |
|----------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sunday..... 28 / 01 / 2024 ..... |       | Yugadanari Power Plant<br>Kerawalapitiya.                                                                                                                                                  |
| DAY                              | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                                                                                  |
| Monday                           | 01/22 | There was a coolend leak in Emergency Diesel Generator radiator. The leak was fixed and tested that is it working properly.                                                                |
| Tuesday                          | 01/23 | There was a problem in blow down drain pit. It's pump was not enough to pump water to the CMB according to the blow down flowrate. To solve that problem flowrate measurements were taken. |
| Wednes day                       | 01/24 | There was a vibration checking programme in main cooling water motors. We joined for that programme and identified about the vibration checking.                                           |
| Thurs day                        | 01/25 | - Poya day -                                                                                                                                                                               |
| Friday                           | 01/26 | Tried to change identify the required pump for the previously mentioned pump in blowdown drain pit.                                                                                        |
| Saturday                         | 01/27 | - Holiday -                                                                                                                                                                                |
| Sunday                           | 01/28 | - Holiday -                                                                                                                                                                                |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

### Vibration monitoring in Cooling Tower motors.

This program was conducted as a condition based monitoring program. The FALCON oIdB device was used to check the vibrations. The FALCON oIdB is a appropriate instrument designed for this purpose, equipped with a range of features to facilitate comprehensive vibration analysis.



The following are some of its features,

- The device is equipped with high precision accelerometers that can detect even the slightest vibrations.
- FALCON o1dB supports multiple channels, enabling simultaneous monitoring of different parts of a machine.
- The device includes Fast Fourier Transform to analyze the frequency components of vibrations. This helps in identifying specific issues like imbalance, misalignment, bearing faults and gear defects.
- It is suitable for on-site monitoring and can be easily transported between different locations within a facility.

Procedure :-

- The triaxial accelerometer and the FALCON o1dB device were first chosen for the application.
- The critical points were determined on the motor where vibration measurements were taken.
- The accelerometer was attached to the predefined monitoring points on the machine. motor.
- Configuring parameters such as sampling rate, measurement duration and frequency range were set up in the FALCON o1dB.
- Vibration measurements were taken.

Chalugana  
SIGNATURE OF TRAINEE

REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC. DIMENSIONS AND SKETCHES TO BE GIVEN WHERE/IF POSSIBLE

DATE : 15/02/2024

P. D. N. Rajesingha (H.E.)  
DESIGNATION AND SIGNATURE



WEEK NO: .....10.....

FOR THE WEEK ENDING

Sunday.....04/02/2024.....

TRAINING LOCATION

Yugadanari Power Plant  
Kerawalapitiya

| DAY        | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                                                                      |
|------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monday     | 01/29 | Monthly maintenance list was prepared. The maintenances was categorized as general maintenance, scheduled maintenance, Operations maintenance and condition based maintenance. |
| Tuesday    | 01/30 | A report was made regarding valve conditioning monitoring using ultrasonic SPT 340 device.                                                                                     |
| Wednes day | 01/31 | Went to the chiller pump house to repair the dosing pump and fixed the pre heater in the tank yard.                                                                            |
| Thurs day  | 02/01 | We had checked the gas turbine stores components and tools about their availability for upcoming main inspection.                                                              |
| Friday     | 02/02 | we had to check the spare parts, their drawing numbers, manufactures, PO numbers, part numbers and made a excel sheet from them about HRSA.                                    |
| Saturday   | 02/03 | - Holiday -                                                                                                                                                                    |
| Sunday     | 02/04 | - Holiday -                                                                                                                                                                    |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

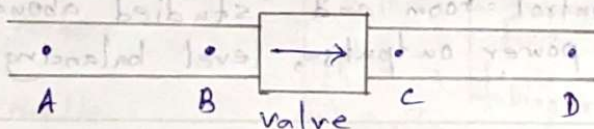
### Valve Condition Monitoring Using Ultrasonic

Valves play a crucial role in the transportation of gas, oil and water. They are used to control the flow, direction, start and stop the flow. So, when valves fail or leak, it will result in the loss of energy or production quality. Whether it's internal or external leaks or issues with flow regulation, valve leaks are a threat to processes. Therefore, ultrasonic



### Leak Detection Method:-

If the sound is louder comparing first downstream (C) with second upstream (B), fluid might be passing through. If the downstream sound level is low, the valve is closed. If the second downstream (D) is louder than first downstream (C), it indicates sound from a source further downstream, suggesting the valve may not be leaking.



Chalangana.  
SIGNATURE OF TRAINEE

REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

DATE : 15/04/2024

DESIGNATION AND SIGNATURE

WEEK NO: .....11.....

FOR THE WEEK ENDING

Sunday..11...../02...../2024.....

TRAINING LOCATION

Yugadanavi Power Plant  
Kerawalapitiya.

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                               |
|-----------|-------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | 02/05 | There were <del>1080</del> more than 1080 items in previously mentioned HRSA items list. So, another few days also we had to check them |
| Tuesday   | 02/06 | "                                                                                                                                       |
| Wednesday | 02/07 | "                                                                                                                                       |
| Thursday  | 02/08 | Visited to the control room and studied about a start up, <del>maxi</del> power outputs, level balancing, synchronizing.                |
| Friday    | 02/09 | Visited to the gas turbine and studied about the compressor part and got an idea about readings of it.                                  |
| Saturday  | 02/10 | - Holiday -                                                                                                                             |
| Sunday    | 02/11 | - Holiday -                                                                                                                             |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

### HRSA Valves Identification.

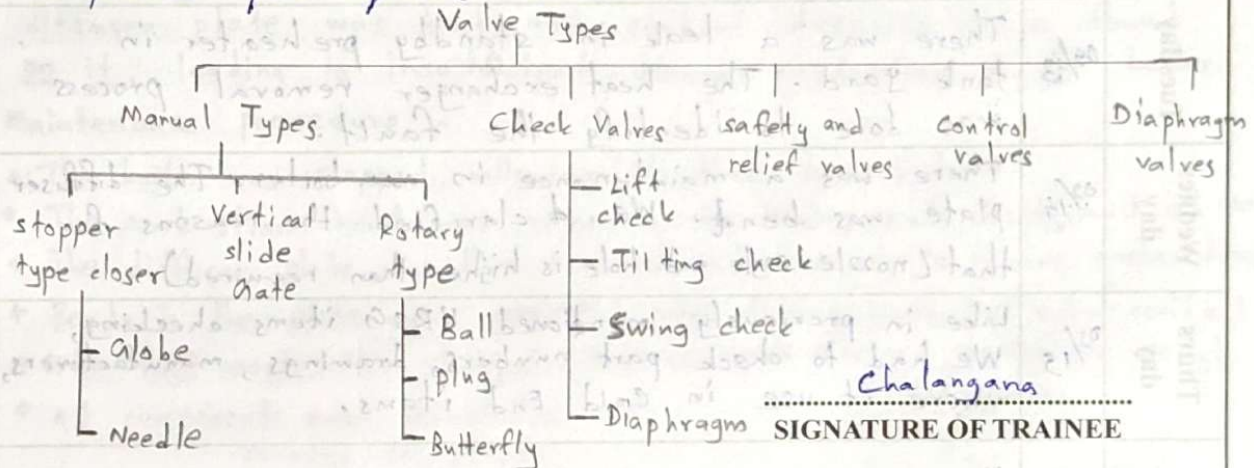
There are many types of valves used in HRSA. They are employed to control flow, direction, start and stop, manage pressure and ensure safety. Given these diverse applications, various types of valves are utilized. Including gate valves, globe valves, check valves, throttle valves, etc.



These valves have different pressure values and varying input and output diameters, which can differ from one valve to another. Therefore, a special numbering method is used to categorize them. For example,

DN150 300# Gate valve.

This indicates a gate valve, with 'DN150' denoting the diameter size of the valve (150 mm or 6 inches) and '300#' representing the maximum operating pressure in pounds per square inch.



#### REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

- Holiday -

- Holiday -

DATE: 15/04/2024

DESIGNATION AND SIGNATURE:

WEEK NO: 12

FOR THE WEEK ENDING

Sunday 18 / 02 / 2024

TRAINING LOCATION

Yugadanavi Power Plant  
Kerawalapitiya

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                                              |
|-----------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | 02/12 | P & ID diagrams of the HRSG were studied and visited there to identify items in the P & ID.                                                            |
| Tuesday   | 02/13 | There was a leak in standby preheater in tank yard. The heat exchanger removal process was done to identify the fault.                                 |
| Wednesday | 02/14 | There was a maintenance in aux. boiler. The diffuser plate was bent. We clarified the reasons for that (nozzle angle and hole is higher than required) |
| Thursday  | 02/15 | Like in previously mentioned HRSG items checking, we had to check part numbers, drawings, manufacturers, where it use in Cold End items.               |
| Friday    | 02/16 | ,                                                                                                                                                      |
| Saturday  | 02/17 | - Holiday -                                                                                                                                            |
| Sunday    | 02/18 | - Holiday -                                                                                                                                            |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

### Maintenance in Aux. Boiler Diffuser.

The Aux Boiler is used to generate steam, which is then utilize to raise the temperature of DM water transferred to the AT, ST, Tank yard and FOTP to achieve the required temperature of that plant. The boiler type is a three-pass wetback smoke tube boiler with a capacity of 10 tons. Its heat output is  $3.37 \times 10^6$  kcal/hr and the pressure is 10.54 kg/cm<sup>2</sup>. HFO is used as fuel.



A maintenance was performed on the burner gun of the boiler. The burner gun comprises a fuel nozzle, ignition system, flame scanner and diffusers. The burner design specifications are as follows.

- Burner type:- Industrial pressure jet • Turn down ratio :- 1:3
- Design standards :- EN 280 • Modulation type:- stepless
- Ignition method :- High voltage spark
- Fuel supply pressure :- 25 kg/cm<sup>2</sup>

The diffuser plate is employed to evenly distribute combustion air throughout the boiler's combustion chamber. However, this diffuser plate was bent, and carbon deposits were found on it, leading to insufficient steam production by the boiler.

#### Maintenance procedure :-

- \* The boiler was stopped, allowing it time to cool down.
- \* The combustion head was removed by loosing nuts, bolts and oil connections.
- \* The diffuser plate was then removed and cleaned to remove carbon deposits.
- \* Bends in the plate were corrected using fitting methods (using anvil and hammer)
- \* The fuel nozzle was changed. (Because nozzle size and angles are wrong).
- \* All components were reinstalled and secured in place.

Chalanganana  
SIGNATURE OF TRAINEE

#### REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED, AND SOLUTIONS, DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

2002 Blower Repair Replacement  
Due to the HRGA burner is operated on the outer  
surface of the HRGA tubes, which can reduce the  
efficiency of the steam generation process. So, 2002  
blower was replaced with 2002 blower.

DATE : 15/04/2024

W. P. D. N. V. Wijesinghe (HE)  
DESIGNATION AND SIGNATURE

WEEK NO: .....13.....

FOR THE WEEK ENDING

Sunday 25 / 02 / 2024

TRAINING LOCATION

Yugadani Power Plant  
Kerawalapitiya.

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                                        |
|-----------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | 02/19 | Safety procedure in Aux. Boiler was checked. The working principal of the air inlet and oil injection methods are understood.                    |
| Tuesday   | 02/20 | We went to take readings in the boiler. Mr. Dammika explained about the safety values and what the time period that we should take the readings. |
| Wednesday | 02/21 | There was a maintenance in HRSG 01. The gaskets and glands are replaced in this maintenance.                                                     |
| Thursday  | 02/22 | There was a vibration checking programme in cooling towers and main cooling pump motors. This was done as a condition base maintenance.          |
| Friday    | 02/23 | - Poya Day -                                                                                                                                     |
| Saturday  | 02/24 | - Holiday -                                                                                                                                      |
| Sunday    | 02/25 | - Holiday -                                                                                                                                      |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

### Soot Blower Gland Replacement.

Due to the flue gas, carbon is deposited on the outer surface of the HRSG tubes, which can reduce the efficiency of the steam generation process. So, soot blowers are used to remove this deposited carbon.



There are three types of soot blowers, Furnace wall soot blowers, stationary soot blowers and long retractable soot blowers. From them, the long retractable soot blower method is used in this power plant.

Gland Replacement procedure:-

- First checked any pressure or steam within the blower was relieved.
- Then nuts and bolts were removed. The hoses and pipes that were connected to the gland were removed.
- The old gland was carefully removed and inspected the surrounding area for any sign of damage.
- Then new gland was replaced into the housing and gasket was placed. Removed pipes, nuts and bolts were also placed.
- Then performed an inspection to ensure that all components were seated properly.

Chalanganah.

SIGNATURE OF TRAINEE

REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

DATE : 13/04/2024

W.P.D.N.N. Wijesinghe (ME)

DESIGNATION AND SIGNATURE

WEEK NO: .....14.....

FOR THE WEEK ENDING

TRAINING LOCATION

Sunday 02 / 03 / 2024

Yugandharvi Power Plant  
Kerawalapitiya.

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                      |
|-----------|-------|--------------------------------------------------------------------------------------------------------------------------------|
| Monday    | 02/02 | There was a maintenance in tube cleaning pump motor in ccw system. The motor was removed and it's bearings were replaced.      |
| Tuesday   | 02/27 | I added some checklists into the plant system. Mainly Cold end and steam turbine preventive maintenance checklists were added. |
| Wednesday | 02/28 | Created an excel sheet about required bearings to the plant.                                                                   |
| Thursday  | 02/29 | I sorted previously made excel according to its bearing numbers. Then I had to study about those many types of bearings.       |
| Friday    | 02/01 | There was an old gas turbine. The parts of that turbine were removed. So I involved it to understand about that gas turbine.   |
| Saturday  | 03/02 | - Holiday -                                                                                                                    |
| Sunday    | 03/03 | - Holiday -                                                                                                                    |

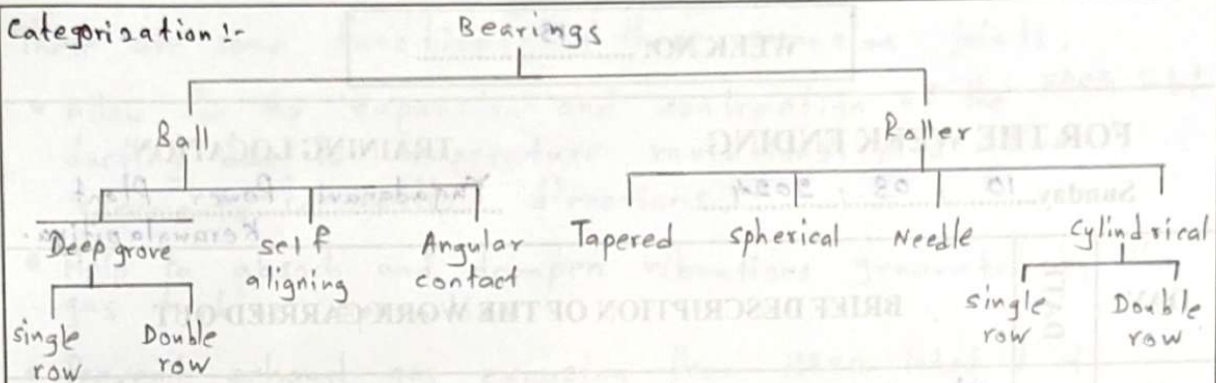
DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

### Bearing Categorization and Number Identification.

A bearing is a machine element that constrains relative motion to only the desired motion and reduces friction between moving parts. A bearing consists of outer ring, inner ring, rolling elements, cage and a seal (optional).



## Category 1:-



## Numbering :-

Bearing numbers provide information about the bearing type, size and other specifications. There can include prefixes, suffixes and numeric codes.

Prefixes often indicate the type or series of the bearing and numeric codes provide information about the bearing size, design and other characteristics. Suffixes are used to denote specific features, modifications, or variants of a standard bearings.

Ex:- 6305 22 C3 → Internal clearance

→ sealing

→ Bore size

→ Bearing series

→ Bearing type.

Chalangana.

SIGNATURE OF TRAINEE

## REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC. DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

HOSE Expansion joint leak resolution

In HOSE assembly, expansion joints are situated at the inlet and outlet. They are typically made from metal and are used to absorb vibration and movement in the system. In this case, the expansion joint at the inlet was leaking. The leak was caused by the joint being loose and the metal being corroded. The joint was tightened and the metal was replaced. The leak was resolved.

DATE : 12/04/2024

W.P.D.N.N. Wijesinghe (ME)  
DESIGNATION AND SIGNATURE

WEEK NO: .....15.....

FOR THE WEEK ENDING

Sunday.....10...../.....08...../.....2024.....

TRAINING LOCATION

Yugadanari Power Plant  
Kerawalapitiya.

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                                                       |
|-----------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | 08/04 | There was a exhaust gas leak in HRSG 02 expansion joint. A bellow was placed on that required area and for further insulation, glasswool was used to cover it.  |
| Tuesday   | 09/05 | I had to check previously made sorted bearing list with a quotation. Evening, I saw previously mentioned turbine was disassembling. I joined it.                |
| Wednesday | 03/06 | The color of the HRSG 01 CRP pump was changed. Therefore, oil was removed and replaced. I had joined it and identified about the gearbox, couplings and motors. |
| Thursday  | 03/07 | There was fault in BT fan 01 in GT 02. The winding was damaged due to a bearing failure. That motor was removed and BT fan 02 bearings also removed.            |
| Friday    | 03/08 | The blowers in ceiling air fan of HRSG 02 were cleaned.                                                                                                         |
| Saturday  | 03/09 | - Holiday -                                                                                                                                                     |
| Sunday    | 03/10 | - Holiday -                                                                                                                                                     |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

### HRSG Expansion joint leak Insulation.

In HRSG assembly, expansion joints (bellows) are situated at the inlet duct. They are typically made from metal or fabric materials and feature a convoluted or corrugated design to provide flexibility while maintaining pressure integrity.



There are some functions of these expansion joints.

- Allow for the expansion and contraction of the HRSG inlet ducting due to temperature variations. Also allow movements in multiple directions.
- Help to absorb and dampen vibrations generated by the gas turbine operation.
- Prevent exhaust gas escaping from HRSG inlet duct.

In HRSG 02, an exhaust gas leak occurred in the expansion joint. The maintenance procedure involved the following steps.

- \* Initially, the leaking areas in the joint were located and identified.
- \* Then, power generation from GT 02 was halted.
- \* The previously identified areas were covered using a new bellow, and nuts and bolts were used to secure it in place.
- \* Further insulation was applied by sealing the ends of the joint with glass wool.
- \* Finally, an inspection was conducted to ensure there were no remaining leaks.

Glass wool is an insulation material type made from recycled glass, which is melted and spun into fine fibers. These fibers are then bound together using a resinous binder to create a light weight and flexible insulation period.

SIGNATURE OF TRAINEE

REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC. DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

Make valve seat grinding using Ventil Orbit to valve grinding machine.

With this valve seat grinding machine, valves and valve seats can be ground. Therefore, valve leakage can be avoided.

DATE: 19/04/21

W.P.D.N.N. Wijesinghe (ME)

DESIGNATION AND SIGNATURE



WEEK NO: ..... 16 .....

FOR THE WEEK ENDING

Sunday..... 17 / 03 / 2024 .....

TRAINING LOCATION

Yugadanari Power Plant  
Kerawalapitiya.

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                                                                                      |
|-----------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | 03/11 | A seat of a gate valve was grinded using newly bought valve grinding machine Ventil Orbit 10.                                                                                                  |
| Tuesday   | 03/12 | Previous day identified that there was a steam leak in HRSG 02. That leak was found in LP preheater header. That small hole was welded using arc welding and non destructive testing was done. |
| Wednesday | 03/13 | I had to check Purchasing Order lists of bearings and also Purchasing Request lists were checked.                                                                                              |
| Thursday  | 03/14 | There was a coupling replacement in condensate booster pump in ST.                                                                                                                             |
| Friday    | 03/15 | I had to identify bearing failures in BT fan motors, ST lube oil pump motor, CT gear box motor, CPP motor. First I found them and checked outer surfaces.                                      |
| Saturday  | 03/16 | - Holiday -                                                                                                                                                                                    |
| Sunday    | 03/17 | - Holiday -                                                                                                                                                                                    |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

Gate valve seat grinding using Ventil Orbit 10 valve grinding machine.

With time, valves and valve seats corrode, hindering proper sealing between valve seats. Therefore, valve leakage occurs. So, if they are worn unevenly or have any imperfections, grinding may be necessary to restore a smooth surface for proper sealing.



For this purpose, the Ventil Orbit 10 valve grinding and lapping machine was utilized. This machine can be used both horizontally and vertically and can accommodate valve diameters ranging from 1.5 inches to 10 inches. The machine operates pneumatically requiring a air pressure of 7 bars.

Maintenance procedure :-

- In this maintenance, a 6 inch gate valve was used. Therefore the vertical arm of the machine was utilized.
- Initially, the valve was cleaned and placed on the table. The vertical arm of the machine was then fixed and 100  $\mu$ m roughness stickers were applied to the grinding surface of machine.
- Next, the machine was fixed onto the valve, and the air supply was connected.
- Starting from a low speed, the procedure was initiated.
- After few minutes, the machine was stopped to inspect the valve seat.
- This procedure was repeated three times using 100  $\mu$ m, 60  $\mu$ m and 30  $\mu$ m roughness stickers respectively.
- Following this, the valve seat was checked for improved sealing, and then the machine was disassembled.

.....  
SIGNATURE OF TRAINEE

REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

DATE : 15/04/2024

.....  
DESIGNATION AND SIGNATURE



WEEK NO: 17

FOR THE WEEK ENDING

Sunday 24 / 03 / 2024

TRAINING LOCATION

Yugadonavi Power Plant  
Kerawalapitiya.

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                                                                                       |
|-----------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | 03/18 | There was a safety valve testing programme in HRSA 01. The set pressure was gradually increased to the design pressure and inspect the working condition.                                       |
| Tuesday   | 03/19 | There was a fire fighting training programme. The reasons for the fire, equipments used to put out the fire and methods were identified.                                                        |
| Wednesday | 03/20 | Bearing failure types were studied and started to prepare a report about it.                                                                                                                    |
| Thursday  | 03/21 | Previously started bearing failure types report was completed. How to find bearing failure types <sup>using visual inspection</sup> , reasons and countermeasures were included in this report. |
| Friday    | 03/22 | A annual PM of main fire water pump as was done. The impeller and outer surfaces were cleaned and gaskets were replaced.                                                                        |
| Saturday  | 03/23 | - Holiday -                                                                                                                                                                                     |
| Sunday    | 03/24 | - Holiday -                                                                                                                                                                                     |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

Identification of Bearing failure types using visual Inspections

Bearing is an essential part that allows machinery to rotate smoothly. When bearing fails, it can cause more friction, vibration and noise. This not only reduce the machine's performance but can also lead to downtime and potential damage to the equipment.



There are many failure types. They can be identified using visual inspections of the bearing rings, cage and rolling elements. Some of main types are,

01. **Wear** :- Occurs due to the friction between rolling elements and raceways.

- causes :- Improper or insufficient lubricant, entry of foreign matter.
- countermeasures :- Reexamine the selection of lubricant or method, improve the sealing.

02. **Fretting** :- Occurs at the interface and between mating surfaces due to slight relative motion induced by vibration, oscillation or cyclic loading.

- causes :- Insufficient interference.
- countermeasures :- Provide greater interference, apply lubricant to the fitting surface.

03. **Creep** :- bearing rings move relative to the shaft or housing.

- causes :- Insufficient interference, insufficient tightening of the sleeve.
- countermeasures :- Provide greater interference, tighten the sleeve as appropriate.

04. **Scoring** :- The formation of shallow grooves or scratches on the surface of bearing components.

- causes :- Poor lubrication, Particles are caught in the surface, shaft bending
- countermeasures :- Apply lubricant to the raceway and rolling surfaces, check the size of the load, Adjust the preload.

05. Rust :- form of corrosion that specifically occurs on iron or steel surfaces when they are exposed to oxygen and moisture.

- Causes :- Contamination by water or corrosive matter, Improper storage condition.
- Countermeasures :- Improve the bearing storage condition  
Improve the sealing device  
Provide rust preventive treatment.

Chalanganana

SIGNATURE OF TRAINEE

REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

DATE : 15/04/2024

W.P.D.N.V. COYAMBATUR (M.E)  
DESIGNATION AND SIGNATURE



WEEK NO: 18

FOR THE WEEK ENDING

TRAINING LOCATION

Sunday 31 / 03 / 2024

Yugadanavi Power Plant  
Kerawalapitiya.

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                                                                                |
|-----------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | 03/25 | Prepared a report about ET II fan motor bearing failure types using visual inspections and gave reasons for them and countermeasures.                                                    |
| Tuesday   | 03/26 | Failure types in condensate recirculation pump drive end bearing were studied and a report was prepared about the reasons for the failure and countermeasures.                           |
| Wednesday | 03/27 | Failure types in CRP motor non-drive end bearing were studied using visual inspections and a report was prepared about the reasons for the failure and countermeasures.                  |
| Thursday  | 03/28 | Failure types in water injection pump motor drive end bearing were studied using visual inspections and a report was prepared about the reasons for the failure and countermeasures.     |
| Friday    | 03/29 | Failure types in water injection pump motor non-drive end bearing were studied using visual inspections and a report was prepared about the reasons for the failure and countermeasures. |
| Saturday  | 03/30 | - Holiday -                                                                                                                                                                              |
| Sunday    | 03/31 | - Holiday -                                                                                                                                                                              |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

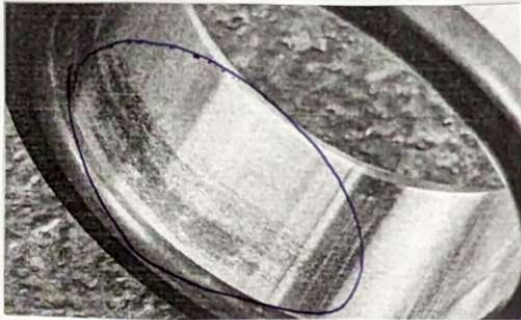
Identification of failure types using visual Inspections in Condensate Recirculation pump bearings.

There were two bearings in CRP. Visual inspections were conducted to identify the failure types.

## Drive End Bearing Failures

### 01. Fretting

#### Inspections :-



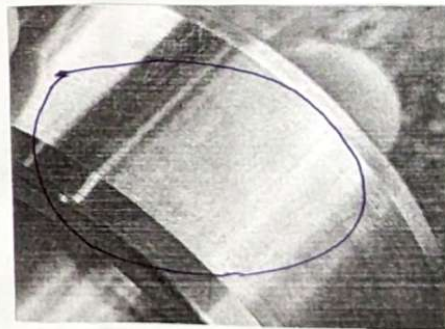
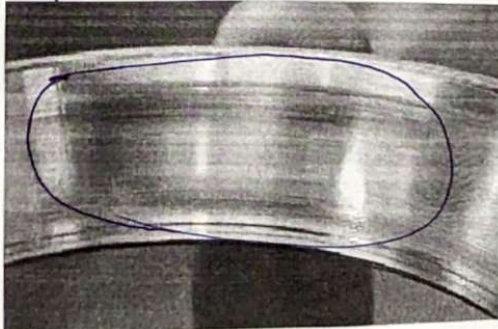
#### Reasons for the failure :-

- Poor lubrication
- Vibration with a small amplitude
- Insufficient interference.

The outer surface of the outer ring and the inner surface of the inner ring <sup>were</sup> exhibited surface damage similar to that observed in fretting conditions.

### 02. Wear

#### Inspections :-



The raceway of the inner ring, the raceway of the outer ring and the rolling elements were exhibited surface damage like that observed in wear conditions.

#### Reasons for the failures :-

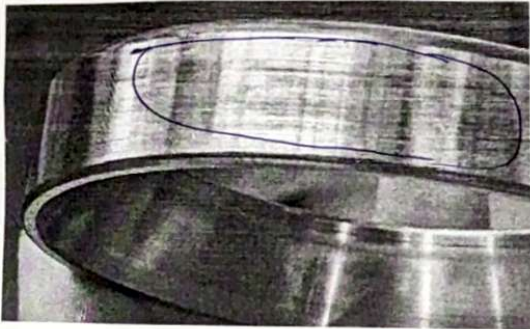
- Entry of foreign matter
- Improper or insufficient lubrication.
- Sliding due to irregular motion of rolling elements.



## Non Drive End Bearing Failure.

### 01. Fretting

#### Inspections :-



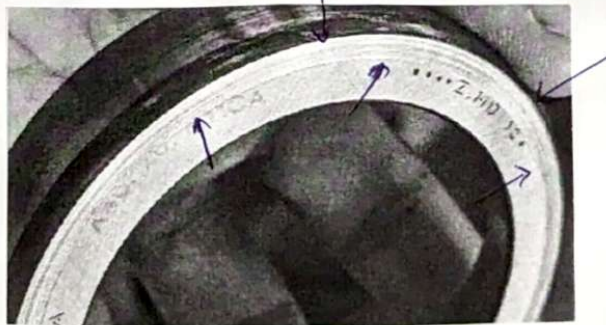
#### Reasons for the failure :-

- Poor lubrication
- Vibration with a small amplitude
- Insufficient interference.

The outer surface of the outer ring and the inner surface of the inner ring were exhibited surface damage like that observed in fretting conditions.

### 02. Scoring

#### Inspections :-



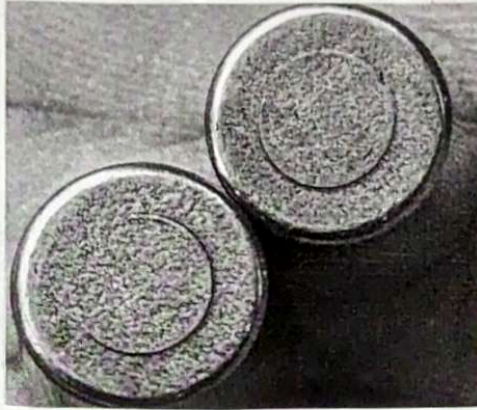
The outer ring and the inner ring were exhibited surface damage like that observed in scoring condition.

#### Reasons for the failure :-

- Excessive load / excessive preload.
- Particles were caught in the surface.
- Poor precision of the shaft and housing.
- Shaft bending.

03. Pitting.

Inspections:-



The rolling elements were exhibited surface damage similar to that observed in pitting condition.

Reasons for the failure:-

- Debris become caught in the lubricant.
- Exposure to moisture in the atmosphere.
- Poor lubrication.

*chelangana*

SIGNATURE OF TRAINEE

REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

DATE : 19/04/2024

*W.P.D. N. N. W. J. Ganghe* (ME)  
DESIGNATION AND SIGNATURE



WEEK NO: 19

## FOR THE WEEK ENDING

Sunday, 07. / 04 / 2024

## TRAINING LOCATION

Yugadanari Power Plant  
Kerawalpitiya.

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                                                        |
|-----------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | 04/01 | There was an electrical lecture about the electrical system in this power plant. Transformers, turbine generator and some motors were discussed in this lecture. |
| Tuesday   | 04/02 | Two separators in FOTP were repaired. Line 04 separator rubber seal of the coupling was damaged and in line 02 separator, the sludge was blocked.                |
| Wednesday | 04/03 | Some checklists were updated to the checklist master in the power plant system. Preventive maintenance were updated using these checklists.                      |
| Thursday  | 04/04 | The belt of the mixed bed blower B was replaced.                                                                                                                 |
| Friday    | 04/05 | The tube cleaning pump non return valve was broken in CCW system. It was prepared.                                                                               |
| Saturday  | 04/06 | - Holiday -                                                                                                                                                      |
| Sunday    | 04/07 | - Holiday -                                                                                                                                                      |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

### Electrical system in Yugadanari Power Plant.

The output power of this Yugadanari Power Plant is 300 MW. That current passes through two using high voltage lines. First one is connected to the GT 01 and the other one is connected to the GT 02 and ST. The output of voltage of the GT and ST

is 14.5 kV. Before distributing, this voltage is stepped up to 220 kV using main transformers. There are two auxiliary transformers to step down 14.5 kV into 6 kV. There are some low voltage transformers used to step down 6 kV into 400 V.

| Main transformers                                                   | Auxiliary transformers           | low voltage transformers. |
|---------------------------------------------------------------------|----------------------------------|---------------------------|
| 2 x 130 MVA rating<br>1 x 140 MVA rating                            | 10 MVA rating                    | 3.5 MVA and 2.5 MVA.      |
| oil natural air natural cooling and oil natural air forced cooling. | Oil natural air natural cooling. |                           |

There are three types of control systems used in this plant.

- Mark III (for gas turbines and steam turbine)
- Distributed control system (for auxiliary plants like WTP, FOTP...)
- PLC (Inside auxiliary plants like soot blowers, caw)

Chalangana.

SIGNATURE OF TRAINEE

#### REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

DATE: 15/04/2024

W.P.D.N.N. Wijesinghe (ME)

DESIGNATION AND SIGNATURE



WEEK NO: 20

## FOR THE WEEK ENDING

Sunday 14 / 04 / 2024

## TRAINING LOCATION

Yugadanavi Power Plant  
Kesawalapitiya

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                            |
|-----------|-------|--------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | 04/08 | Thermal photos were taken of HRSG 02 were taken to identify the exhaust leaks. Evening, there was a lecture about auxiliary boilers. |
| Tuesday   | 04/09 | Instrument air leak was occurred in a pipe in WTP sea water flow line. It was repaired and checked.                                  |
| Wednesday | 04/10 | - New year holidays -                                                                                                                |
| Thursday  | 04/11 | - New year holidays -                                                                                                                |
| Friday    | 04/12 | - New year holidays -                                                                                                                |
| Saturday  | 04/13 | - New year holidays -                                                                                                                |
| Sunday    | 04/14 | - New year holidays -                                                                                                                |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

### Auxiliary Boiler and Hot water system.

This plant features two auxiliary boilers, both branded SHELLMAX, each with a capacity of 10 tons. These boilers are three pass smoke type fire tube boilers, utilizing HFO as fuel. Each boiler has a heat output of  $5.39 \times 10^6$  kCal/h and operates at a pressure of

approximately 8 bars, with internal temperatures ranging from 450 to 500 Celsius degrees.

In the combustion chamber, heat is generated by oxidizing fuel, and the necessary heating surfaces absorb this heat into water, converting it into steam. The steam is then separated from the water by a steam separation area on the water's surface. Two-inch tubes are used to transfer the exhaust gas, providing sufficient surface area for high absorption efficiency. The heat from the exhaust gas is used to increase the inlet air temperature.

The burner programmer motor, relays and cam operated contacts ensure the combustion chamber is purged before the fuel is sprayed. The inlet air dampers are automatically controlled by cam operated contacts according to the fuel spray requirements.

Untreated HFO is stored in a day tank and pumped through the HFO header ( $P = 3-4$  bars,  $T = 90^{\circ}\text{C} - 100^{\circ}\text{C}$ ) using two ring main pumps. Two HFO preheaters then increase the HFO temperature to 120 Celsius degrees. The HFO is pressurized to 20 bars before being used by the burner.

In steam-water side, DM water is first pumped into the feed water tank using a makeup water pump where the  $T$  is around 100 Celsius degrees. Each boiler has two feed water pumps that pump this water into the boiler, where the internal pressure is about 10 bars and the  $T$  is around 100 Celsius degrees. The generated steam is directed through the steam header ( $P = 7-8$  bars,  $T = 180^{\circ}\text{C}$ ). Two heat exchangers use this steam to heat water, raising its



temperature to  $150 - 160^{\circ}\text{C}$ . The condensed steam returns to the feed water tank.

Steam is used in the gland sealing and the Ejector super heaters. This steam is also produced by the auxiliary boiler system. An electric heater is used in the auxiliary boiler and it is increased the steam T according to the gland sealing steam and Ejector steam. In a startup of the auxiliary boilers, this electric heater is used to increase the temperature of the HFO.

Gland sealing pressure = 7 bars

Temperature =  $300^{\circ}\text{C}$ .

Ejector steam pressure = 6.7 bars

Temperature =  $240^{\circ}\text{C}$ .

Chalangana.

SIGNATURE OF TRAINEE

#### REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

DATE : 10/03/2024

W.P.D.N.V. W. P. D. N. V. (ME)  
DESIGNATION AND SIGNATURE

WEEK NO: 21

## FOR THE WEEK ENDING

Sunday 21 / 04 / 2024

## TRAINING LOCATION

Yugadagari Power Plant  
Kerawalapitiya.

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                                                                                    |
|-----------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | 04/15 | - New Year Holidays -                                                                                                                                                                        |
| Tuesday   | 04/16 | - New Year Holidays -                                                                                                                                                                        |
| Wednesday | 04/17 | Main fuel pump (HFO) and flow divider in Aux. Compartment of GT 02 was removed to place a new one.                                                                                           |
| Thursday  | 04/18 | The auxiliary shaft in aux. compartment, the atomizing air booster pump and 14 nozzles in GT 02 are removed to check defects and replace required items.                                     |
| Friday    | 04/19 | Steam turbine main inspection was started with removing insulations in ST. Measured Bull hole thicknesses in GT 02 combustion chamber were measured using vernier caliper.                   |
| Saturday  | 04/20 | Upper casing of the GT 02 turbine part was removed to replace inside buckets, nozzles and other parts.                                                                                       |
| Sunday    | 04/21 | Coupling between generator and the turbine was started to disengage. To removed coupling bolts, hydraulic pressure machine was used. ST stage, clearance values of the springs were checked. |

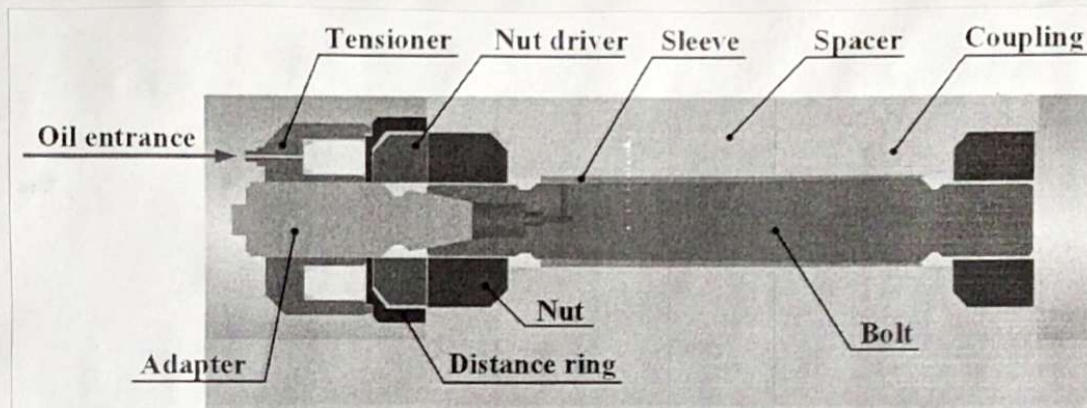
DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

Bolts removing of the coupling between generator shaft and turbine shaft using hydraulic pressure pump.

Coupling is a essential mechanical device that transmits rotational motion between shafts. Hydraulic coupling bolts are commonly used in rigid coupling systems.



When using a regular bolt to secure a flange, a gap between the bolt and the flange hole can cause the bolt to loosen over time. To prevent this, hydraulic coupling bolts have sleeves that fill the gap between the bolt and the flange hole. The inner surface of the sleeve and the contact surface of the bolt are designed with a specific slope to ensure a secure fit.



Nut removing process

Procedure :-

- The tensioner was threaded on by a hand ratchet.
- The high pressure hose line had been coupled to the hydraulic pump and hydraulic pressure was applied to the tensioner.
- The nut was loosened using a spring-loaded gear drive system with a hand torque wrench.
- The pressure was then released from the bolt tensioning cylinder and the tensioner automatically returned to its starting position by means of an internal spring return system.

- 
- A photograph of a mechanical assembly, likely a turbine or compressor section. The assembly consists of a central shaft with a nut and washer on the left, and a series of blades or vanes on the right. Hand-drawn arrows indicate the flow of air or gas from the left, through the blades, and out to the right.

oil

Bolt/stud

## Chalangana

REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

.....  
 W. P. D. N. N. Wijesinghe (ME)  
 DESIGNATION AND SIGNATURE



WEEK NO: 22

FOR THE WEEK ENDING

Sunday 28 / 04 / 2024

TRAINING LOCATION

Tugadanari Power Plant  
Kerawalapitiya.

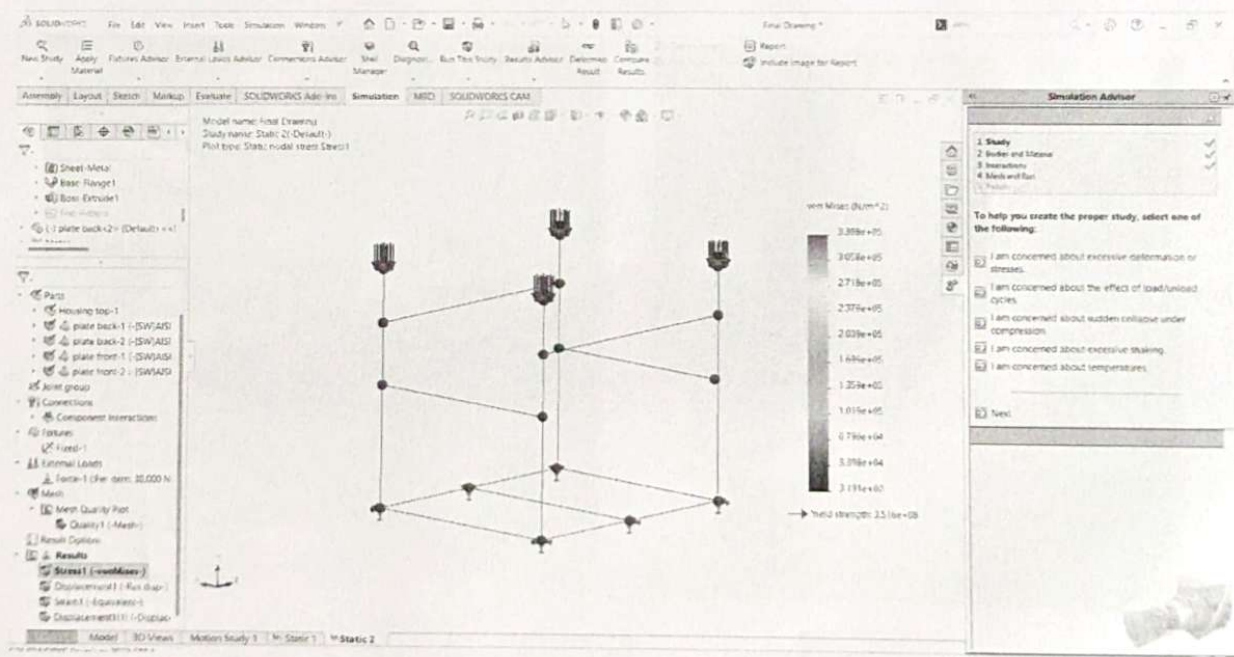
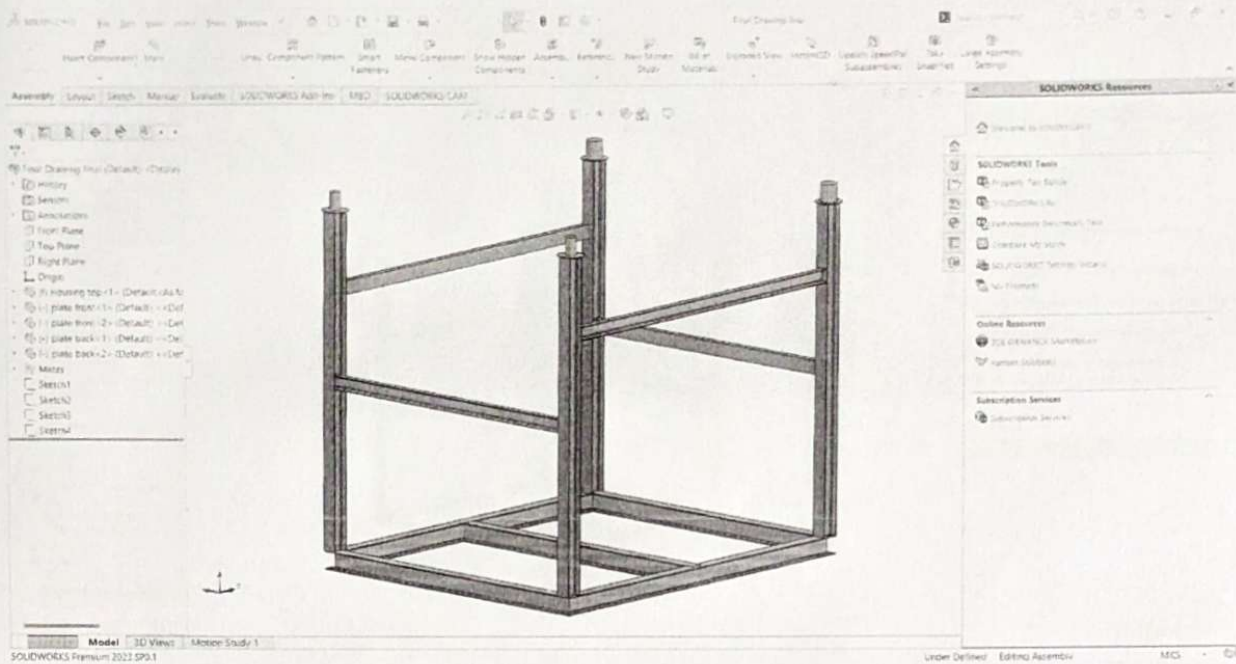
| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                                                                                               |
|-----------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | 04/22 | A structural support to hold the upper casing of the LP and HP steam turbine was designed using solidworks according to the required dimensions.                                                        |
| Tuesday   | 04/23 | Fabrication of the previously designed solidworks design was done using I beams and C channels. Fabricators were helped to successfully complete this task.                                             |
| Wednesday | 04/24 | CCW pump 2 was removed to replace the bearings. The two journal bearings in generator side rotor were removed to check defects of them.                                                                 |
| Thursday  | 04/25 | A report about cracks inspection in GT combustion chamber can casing and corrosion inspection in casing of the turbine was done. The upper casing of the ST high pressure and low pressure was removed. |
| Friday    | 04/26 | Ultrasonic testings for the tilted pads in journal bearings were done to find cracks or any other defects in the bearing. Dry penetration tests were done for the NDE fan blades in the generator.      |
| Saturday  | 04/27 | The clearance values of the between labyrinth seals and between rotor end point to the pedestal were taken using filler gauges and micrometer bore gauge.                                               |
| Sunday    | 04/28 | The turning gear and the gear box of the steam turbine were removed for the inspection and testings.                                                                                                    |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

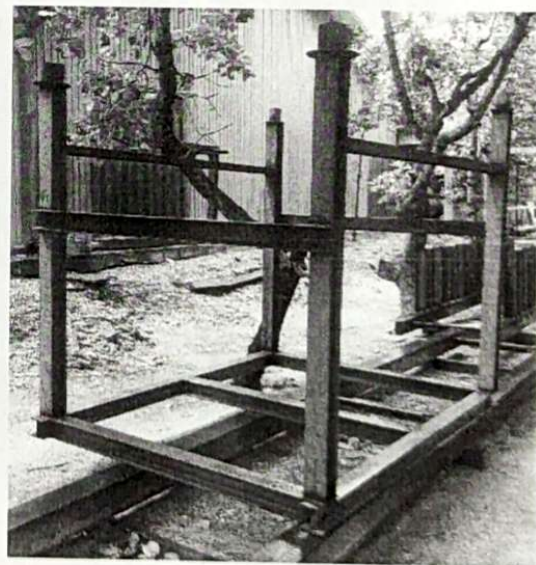
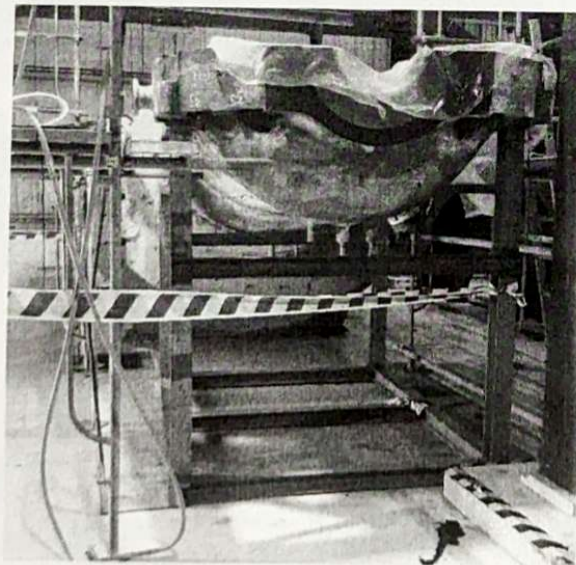
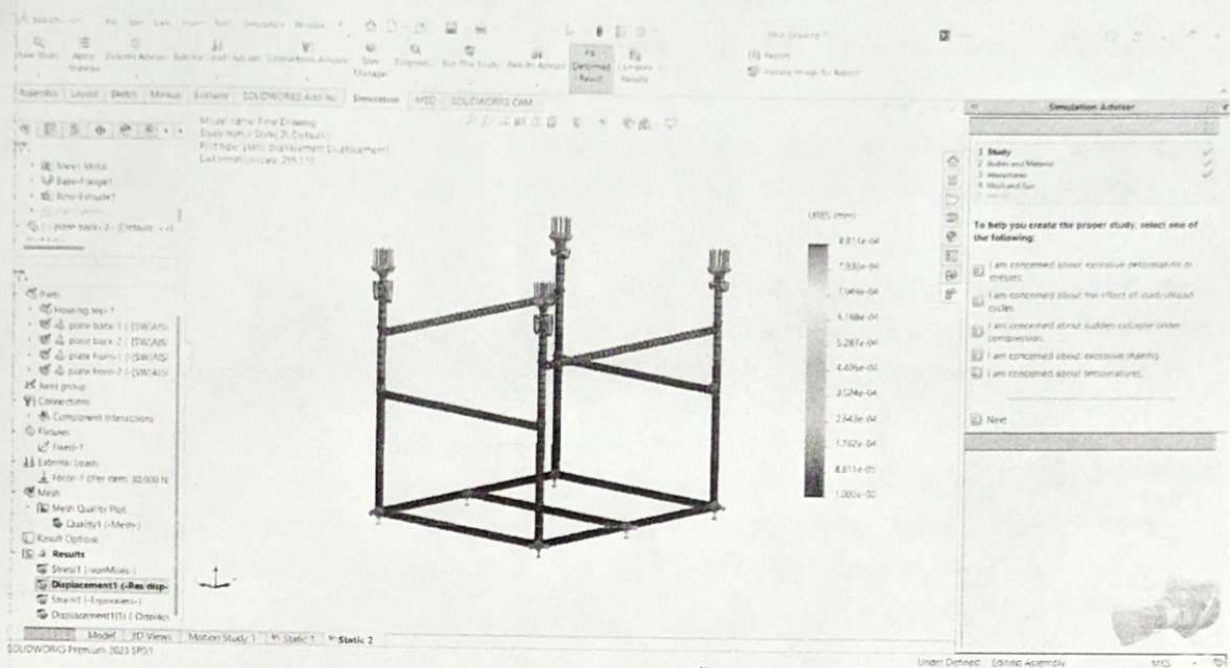
Designing and fabricating of the A structure for the ST upper casing.

This structure was designed to place the upper casing of the steam turbine.

- First the design idea was discussed with the engineers and the required measurements were taken using measuring tape and level meter.







Final Product.

- The a 2D design was sketched in a paper to get an idea.
- According to the previous sketches, A 3D design was designed using SOLIDWORKS software.
- For the structure, 100 mm size I beams and C channels were used due to their high strength and the availability as scrap parts in Yagadanari Power Plant.
- C channels were used for the cross beams because there were no horizontal force. Therefore, the strength in horizontal direction was not essential. They were used as supports for the columns.
- A simulation was done for the whole structure using SOLIDWORKS software with the safety factor of 2. The used force was 60000 N according to the weight of the upper casing.
- The maximum stress was  $2.398 \times 10^5 \text{ N/mm}^2$  and that is less than the yield strength  $3.51 \times 10^8 \text{ N/mm}^2$ . The maximum displacement was 0.000881 mm on the top plates. Therefore, the structure was stable according to the design.
- Finally, fabrication was done with the help of fabricators in this plant.
- Mig welding was used to the fabrication due to its higher welding strength.

Chalangana

SIGNATURE OF TRAINEE

#### REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

DATE: 12/02/2024

W. P. D. N. N. Wijesinghe (ME)  
DESIGNATION AND SIGNATURE



WEEK NO: 23

FOR THE WEEK ENDING

Sunday 05 / 05 / 2024

TRAINING LOCATION

Kugadanari Power Plant  
Kerawalapitiya.

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                              |
|-----------|-------|------------------------------------------------------------------------------------------------------------------------|
| Monday    | 04/09 | The rotor of the steam turbine was removed with the buckets and it was placed for the inspections and blastings.       |
| Tuesday   | 04/30 | The HRSG tubes and soot blowers were inspected after visiting inside to the HRSG 02. This HRSG 02 was being cleaned.   |
| Wednesday | 05/01 | The surface roughness values in each bucket of the rotor in steam turbine were measured using surface roughness gauge. |
| Thursday  | 05/02 | New buckets were being placed in AT 02 turbine and the top casing of the steam turbine was being cleaned.              |
| Friday    | 05/03 | Blue match test was done for the steam turbine casing.                                                                 |
| Saturday  | 05/04 | Bore alignment for the steam turbine casing was done using a guitar string.                                            |
| Sunday    | 05/05 | - Leave -                                                                                                              |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

Surface profile visual and quantitative analysis in Steam Turbine Rotor Blades.

This roughness analysis was performed following ASTM D4417, a global standard for measuring the surface profile of blast cleaned steel. A combination of methods surface comparator, Electronic needle gauging and Replica tape were used to match the curvatures of the blading and ensure the highest possible accuracy.

### Surface Comparator :-

Comparator is a flat, steel plate which contains a number of reference surface profiles, formed on a corrosion resistant metal. These reference profiles are then compared to the surface. There were five reference classes.

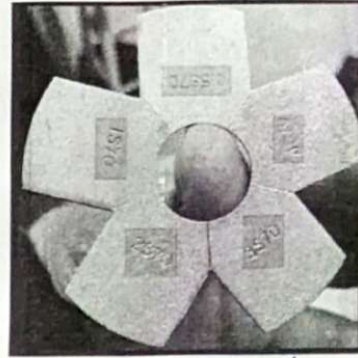
0.5570 = 1.9 microns

1570 = 21.2 microns

2570 = 25.8 microns

3370 = 37.7 microns

4570 = 47.5 microns.



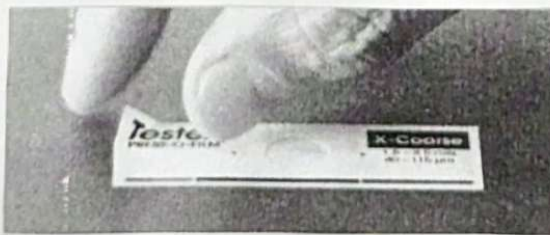
Surface Comparator

### Electronic Needle Gauge :-

Eleometer 934 T surface profile gauge was used as the needle gauge. Peak to valley surface profile heights were measured and recorded from this device. The tip angle was 60 degrees with conical shape. The device was calibrated using a glass plate and then used. Due to the shape of the impulse type of turbine buckets, this device was not used for the HP side.

### Replica Tape method :-

First the tape was placed on the surface and burnished it into the surface. Then the tape was removed and measured by using an electronic needle gauge. This method was used for the places where electronic needle gauge can't be used due to the shapes of the various buckets.



Replica tape



Electronic Needle Gauge



## Results :-

The net value distribution over the high-pressure blades were around 30-40 microns, and the surface comparator showed similarity to 1570. This profile remains the same for the steam addition side face and exit face regions, while the values similarly to 2570 were showed by blades closer to the low pressure segment on the exit side.

The low pressure side was measured mainly using the electronic needle gauge. The pre analysis was done by Electro Elcometer 224T placing right over the uniform surfaces on the inner side of blades.  $60\mu$  for the max,  $\sim 6\mu$  for the min and  $8.16\mu$  for the mean were the results, and for the exit sides of blades  $367\mu$  for the max,  $\sim 7\mu$  for the min and  $90.42\mu$  for the mean were the results.

Therefore, there were minor rust or stains over the high pressure segment while relatively greater observed at the low pressure segment which is near Chalargana to the condensers.

SIGNATURE OF TRAINEE

## REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

DATE : 12/05/2024

P. D. W. N. Wijesinghe (M.E)  
DESIGNATION AND SIGNATURE



WEEK NO: 24

| FOR THE WEEK ENDING   |       | TRAINING LOCATION                                                                                                                                                                                       |
|-----------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sunday 12 / 05 / 2024 |       | Yugadanavi Power Plant<br>Kerawalapitiga.                                                                                                                                                               |
| DAY                   | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                                                                                               |
| Monday                | 05/06 | Identification of vane body pitting, grooving, erosion and erosion in tenons were done for the random buckets in each stage of the rotor.                                                               |
| Tuesday               | 05/07 | Aluminum oxide blasting was done for the diaphragms in steam turbine. Arit size 180 Aluminum oxide was used for that.                                                                                   |
| Wednes day            | 05/08 | New fuel nozzles for the AT 02 combustion chamber were placed.                                                                                                                                          |
| Thurs day             | 05/09 | The width of the buckets of in steam turbine 20th stage were measured using a depth gauge to identify about the erosion of the buckets.                                                                 |
| Friday                | 05/10 | The steam turbine rotor was placed inside the turbine to set the clearances in HP side diaphragms. Bolts were used to set the clearances.                                                               |
| Saturday              | 05/11 | The thickness values of the tube bundles and the headers of LP economizer, evaporator and <del>sp</del> superheater and HP economizer were measured using Olympus 38DL PLUS ultrasonic thickness gauge. |
| Sunday                | 05/12 | The thicknesses in HP evaporator, superheater headers and tube bundles in HRSA 02 were measured using previously used Olympus 38DL ultrasonic thickness gauge.                                          |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

### Steam Turbine Rotor Buckets Inspection.

Proper maintenance and inspection of turbine rotors and buckets primarily enhance efficiency. Turbines can suffer from water erosion, solid particle erosion, or chemical attacks during operation. This can degrade the quality of the turbine. By implementing procedures to monitor the unit's condition during

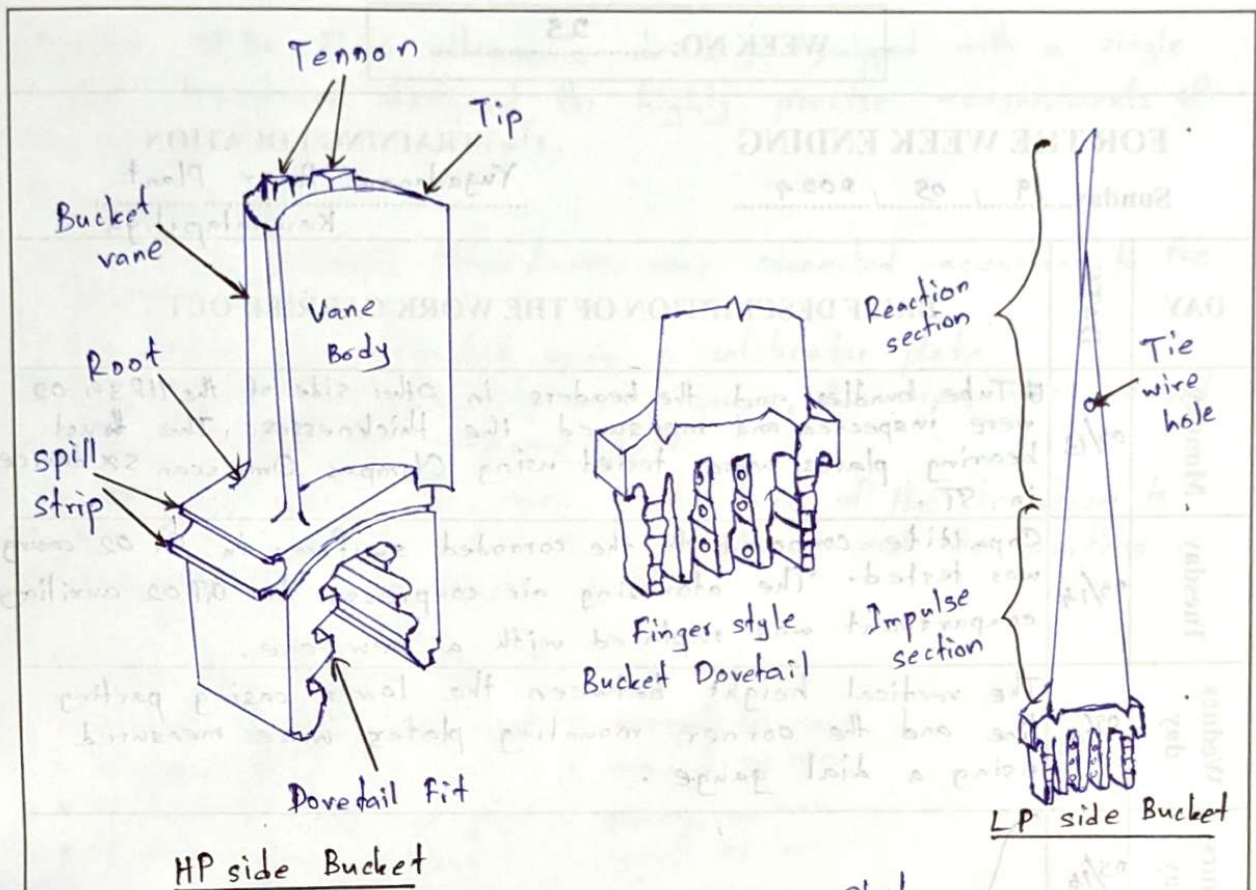


operation and performing through inspections and maintenance during major outages, this can be significantly reduced. Consequently, an inspection was carried out to identify various types of defects in the rotor buckets. The following defects were the mainly focused in the inspection.

- Tenon erosion.
- Cover grooving, rubbing, lifting, curling and mechanical damage.
- Tearing or thinning of vane edges
- Pitting of grooving of vane body.
- Lifting of dovetails.
- Looseness of dovetail pins or keys.
- Erosion of vane section.
- Cracks in tie wire hole.

According to the inspections,

The vane edges in every bucket of the 20<sup>th</sup> stage (both admission and exit) were eroded. The vane bodies were slightly eroded. The vane edges in every bucket of the 19<sup>th</sup> stage were also slightly eroded. There were no cracks in the tie wire holes. The dovetail pins and keys fitted very well, so there was no lifting of dovetails or looseness of dovetail pins or keys. There were small tenon erosion and mechanical damages in few stages. Pitting and grooving of the vane bodies were identified in the HP side stages 1 to 5.



HP side Bucket

LP side Bucket

Chalangana  
SIGNATURE OF TRAINEE

REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

LP side upper diaphragm was attached to the steam turbine and then the upper casing was placed and tightened using heating bolts.

The casing clearance in main shaft off control valve in HP was measured using a dial gauge and the value was 0.12 mm. A bore scope was used to check the & sealing area.

DETAIL - END NOTES ON WORK CARRIED OUT PROBLEMS ENCOUNTERED AND HOW THEY WERE OVERCOME AND SKETCHES TO BE GIVEN WHERE/WHEN POSSIBLE

Measuring the thickness of the drums, tube bundles and headers in HP & LP.

The HP & LP drums of 100 tons each were lifted with the use of 200 tons capacity lift. The lifting was done by 400 mm diameter cables and shackles and was done in a very safe manner.

DATE: 20/05/21

P. D. N. N. Wijesinghe (ME)  
DESIGNATION AND SIGNATURE



WEEK NO: 25

FOR THE WEEK ENDING

Sunday 19 / 05 / 2024

TRAINING LOCATION

Yugadanavi Power Plant  
Kerawalapitiya

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                                                                                             |
|-----------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | 05/13 | Tube bundles and the headers in other side of the HRSG O2 were inspected and measured the thicknesses. The thrust bearing plates were tested using Olympus Omniscan SX device in ST.                  |
| Tuesday   | 05/14 | Copaltite compound for the corroded surfaces in AT O2 casing was tested. The atomizing air compressor in AT O2 auxiliary compartment was replaced with a new one.                                     |
| Wednesday | 05/15 | The vertical height between the lower casing parting line and the corner mounting plates were measured using a dial gauge.                                                                            |
| Thursday  | 05/16 | Leave                                                                                                                                                                                                 |
| Friday    | 05/17 |                                                                                                                                                                                                       |
| Saturday  | 05/18 | LP side upper diaphragms were attached to the steam turbine and then the upper casing was placed and tightened using heating bolts.                                                                   |
| Sunday    | 05/19 | The seating clearance in main shut off control valve in HP was measured using a dial gauge and the value was 4mm. But the required was less than 2mm. A borescope was used to check the seating area. |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

Measuring the thickness of the drums, tube bundles and headers in HRSG O2.

The HRSG consists of more than 100 tube bundles with headers. The primary objective of this program was to assess whether the drums, tubes and headers had experienced any wear over time. Based on a previously created graph, thickness measurements were taken at random using the

Olympus 38 DL PLUS ultrasonic device, equipped with a single element transducer designed for highly precise measurements of thin or multilayer materials.

**Procedure :-**

- The single element transducer was connected according to the requirement.
- The device was calibrated using a calibrator plate.
- According to the previously made graph, the areas were marked before taking the measurements.
- Water and grease were used to the face of the transducer to remove air gap between the transducer and the surface.
- Measurements were taken.

Following are the average thickness values,

- HP drum hemispherical part = around 60 mm
- HP drum cylindrical part = around 90 mm
- LP drum hemispherical part = around 95 mm
- LP drum cylindrical part = around 95 mm
- LP side tubes = 3-4 mm
- LP side headers = 19 - 13.5 mm
- HP side tubes = 3-5 mm
- HP side headers = 20, 25 and 30 mm.

Cholangana

SIGNATURE OF TRAINEE

**REMARKS AND CERTIFICATION BY THE ENGINEER / T.O**

DATE: 30/06/24

DESIGNATION AND SIGNATURE



WEEK NO: 26

FOR THE WEEK ENDING

Sunday 26 / 05 / 2024

TRAINING LOCATION

Yugadanani Power Plant

Keravalapitiya

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                                                        |
|-----------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | 05/20 | The interference between bearing of the turbine and its housing was measured using led (Pb) wires and the installation of the bearings was done. |
| Tuesday   | 05/21 | - Leave -                                                                                                                                        |
| Wednesday | 05/22 | Previously repaired sea water bed pump was place in the intake of the seawater line.                                                             |
| Thursday  | 05/23 | The rotor in generator was placed after checking the faults and repairing the required parts.                                                    |
| Friday    | 05/24 | The Mew pump had been removed to change the bearings and the impeller. It was again fixed with the motor.                                        |
| Saturday  | 05/25 | The turning gear motor with the governor motor were again installed in the steam turbine.                                                        |
| Sunday    | 05/26 | Shaft alignment program was started in cooling towers. Easy Laser XT600 device was used for the program.                                         |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE

Shaft alignment program in Cooling Tower motors using EASELASER XT600 device.

Proper shaft alignment reduces the risk of mechanical failures, increases efficiency and lowers operational costs. Therefore, a shaft alignment program was conducted for the cooling tower fans, specifically for the shaft

between the motor and the gearbox. The EASYLASER XT770 device was used for this program.

The EASYLASER XT~~660~~770 is a laser alignment system designed for industrial applications. It is utilized primarily for the precision alignment of machinery, ensuring optimal performance and reducing wear and tear. This device can measure shaft alignment, straightness, flatness and more making it versatile for various alignment tasks. This device is equipped with wireless technology and allows for easy data transfer and remote monitoring.

#### Procedure :-

- First, we ensured that all machines were stopped according to safety protocols.
- Laser transmitters and detectors were mounted on the shaft using chain brackets.
- The verti. horizontal shaft alignment program was selected from the device's interface and the dimensions such as the distance between the measurement points were entered.
- Initial measurements were taken while rotating the shaft manually. The data were captured at specific points, typically at 9, 12, 3 o' clock.
- The current misalignment was calculated automatically by the device after collecting data from each point.



- The position of the motor was adjusted based on the alignment data. Shims under the motor feet were used to change the position of the motor in vertical plane. This was done by adding and removing shims with various thicknesses.
- The horizontal plane alignment was done by moving the motor in horizontal plane. Eight bolts were used for that.
- After the adjustment, the alignment was re-measured and improvements were checked.
- This measurement and adjustment process was done until the shafts were aligned within acceptable tolerances.

Chalangana.

SIGNATURE OF TRAINEE

REMARKS AND CERTIFICATION BY THE ENGINEER / T.O

DATE : 30/05/24

W. P. D. N. N. Wijesinghe (ME)  
DESIGNATION AND SIGNATURE

WEEK NO: 07

FOR THE WEEK ENDING

Sunday 02 / 06 / 2024

TRAINING LOCATION

Yugadanari Power Plant  
Kerawalapitiga

| DAY       | DATE  | BRIEF DESCRIPTION OF THE WORK CARRIED OUT                                                                      |
|-----------|-------|----------------------------------------------------------------------------------------------------------------|
| Monday    | 05/07 | The instrument air line to the GT and ST generators was repaired according to previously identified leaks.     |
| Tuesday   | 05/08 | HRSA washing was started. Dry Ice blasting in HRSA 02 tube bundles was completed and then started to clean it. |
| Wednesday | 05/29 |                                                                                                                |
| Thursday  | 05/30 |                                                                                                                |
| Friday    |       |                                                                                                                |
| Saturday  |       |                                                                                                                |
| Sunday    |       |                                                                                                                |

DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., DIMENSIONS AND SKETCHES TO BE GIVEN WHEREVER POSSIBLE





FOR USE BY NAITA OFFICIALS ONLY

**INSPECTION REPORT**

FOR THE WEEK ENDING

TRAINING LOCATION

NAME AND DESIGNATION OF OFFICER :

DATE

REMARKS :

SIGNATURE

NAME AND DESIGNATION OF OFFICER :

DATE

REMARKS :

SIGNATURE

NAME AND DESIGNATION OF OFFICER :

DATE

REMARKS :

SIGNATURE

NAME AND DESIGNATION OF OFFICER :

DATE

REMARKS :

SIGNATURE



REMARKS AND ADDITIONAL NOTES:

| INSPECTION REPORT                 |             |
|-----------------------------------|-------------|
| NAME AND DESIGNATION OF OFFICER : | DATE :      |
| REMARKS :                         |             |
| SIGNATURE :                       | SIGNATURE : |
| NAME AND DESIGNATION OF OFFICER : | DATE :      |
| REMARKS :                         |             |
| SIGNATURE :                       | SIGNATURE : |
| NAME AND DESIGNATION OF OFFICER : | DATE :      |
| REMARKS :                         |             |
| SIGNATURE :                       | SIGNATURE : |
| NAME AND DESIGNATION OF OFFICER : | DATE :      |
| REMARKS :                         |             |
| SIGNATURE :                       | SIGNATURE : |

## PROGRESS REPORT OF AN ESTABLISHMENT ON TRAINING

### PERFORMANCE

Name of Establishment : Lakdhanavi Limited.

Period of Training - From : 27/11/2023 To : 30/03/2024

Comments of Training :

1. Conduct Good

2. Attitude to work Good

3. Attendance Good

NO. OF DAYS LEAVE TAKEN

AUTHORIZED

UNAUTHORIZED

03

03

-

Signature of the officer incharge for training  
(with rubber stamp)

Lakdhanavi Limited

300MW CCPP

No. 90, Muthurajawela Industrial Zone  
Kelavani, India

## PROGRESS REPORT OF AN ESTABLISHMENT ON TRAINING

### PERFORMANCE

Name of Establishment : .....

Period of Training - From : ..... To : .....

Comments of Training :

1. Conduct

2. Attitude to work

3. Attendance

NO. OF DAYS LEAVE TAKEN

AUTHORIZED

UNAUTHORIZED

Signature of the officer incharge for training  
(with rubber stamp)